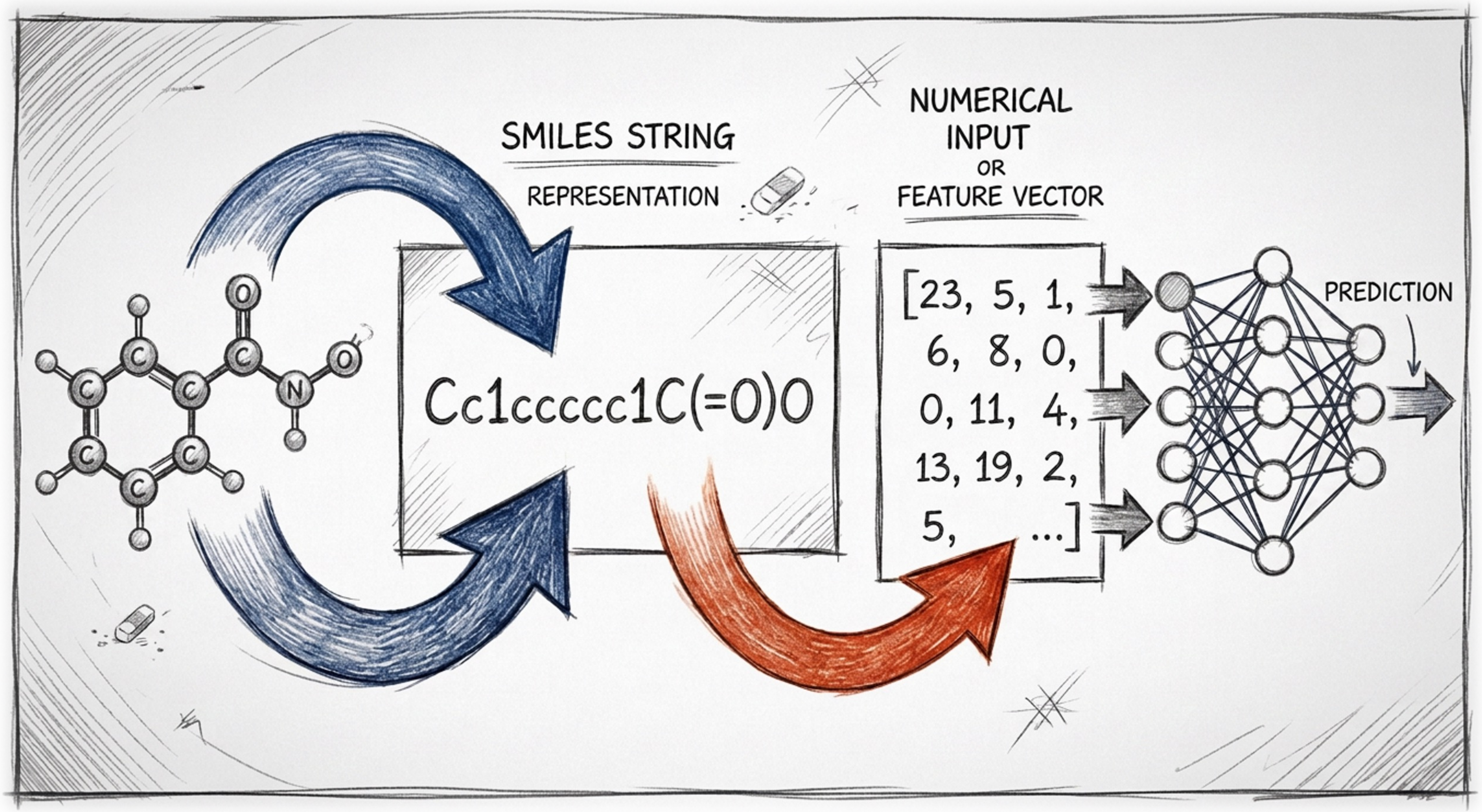
Chemical Representations for Machine Learning



J. Schleinitz · Monday, August 10, 2026 · RSC 275, Caltech

Plan

- **Evolution of "data-driven" chemistry**
- Overview of existing representations
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- Data visualization

Evolution of Data-Driven Chemistry

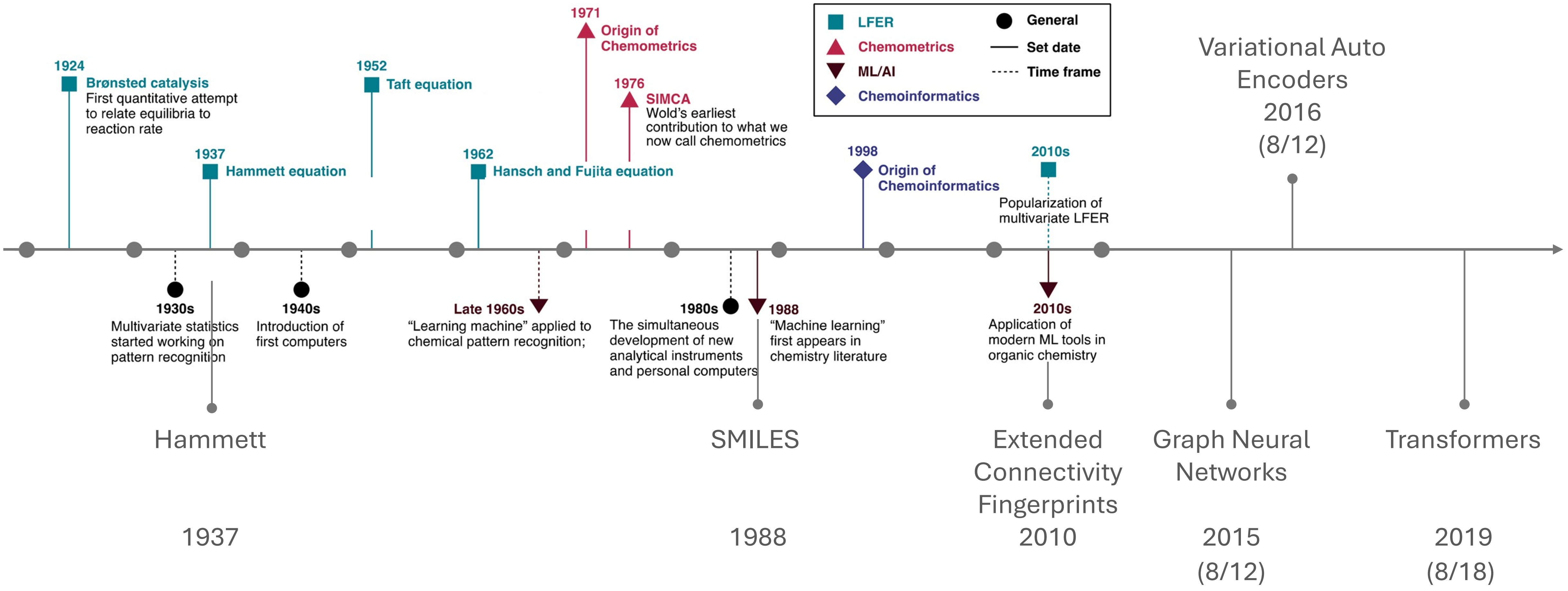
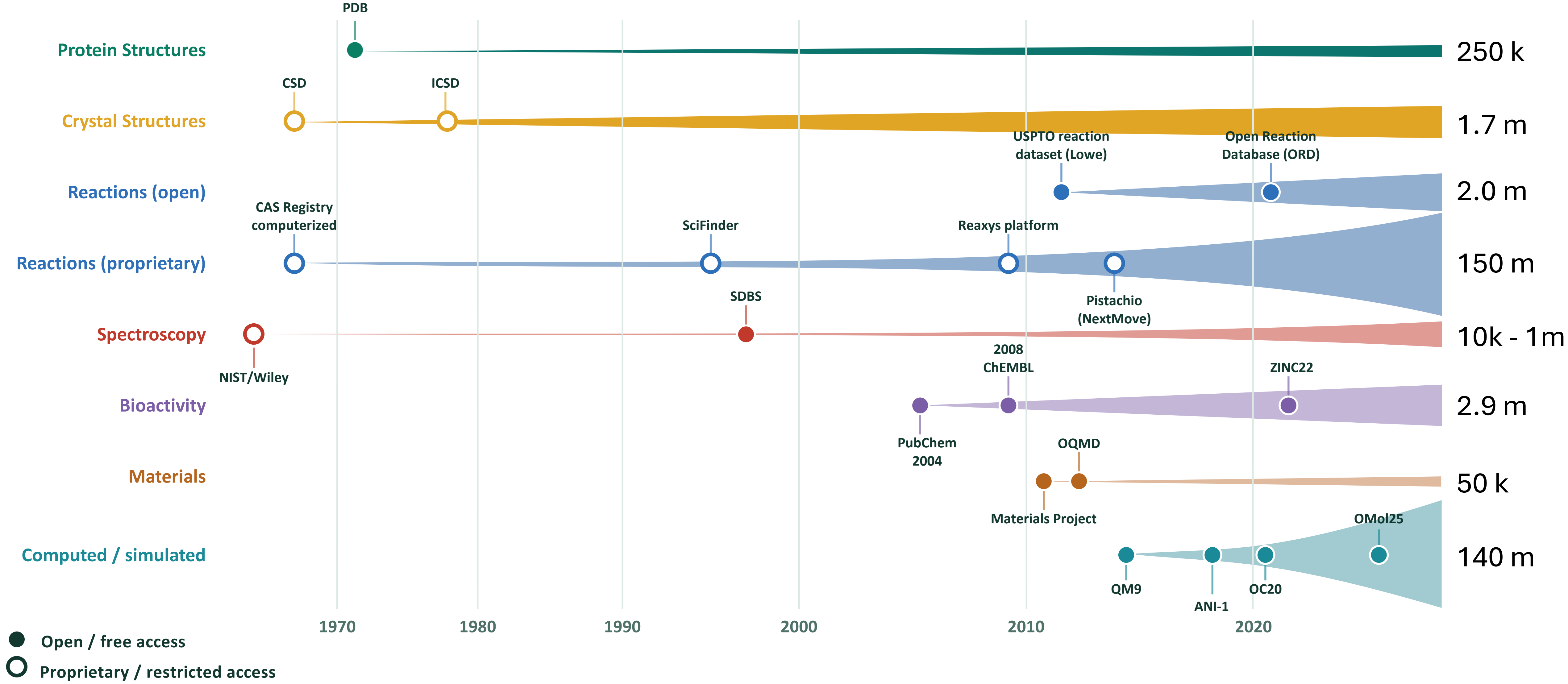
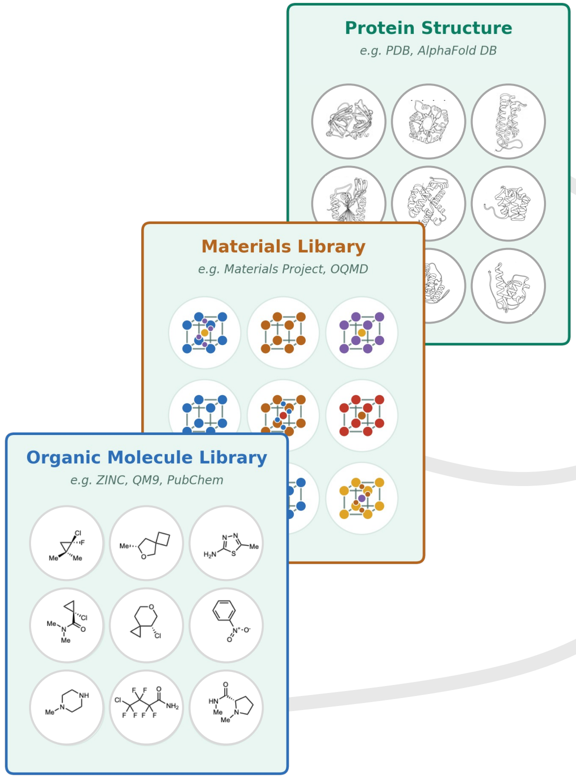


Figure adapted from Williams, W. L.; Zeng, L.; Gensch, T.; Sigman, M. S.; Doyle, A. G.; Anslyn, E. V. *ACS Cent. Sci.* **2021**, 7 (10), 1622–1637

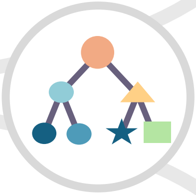
History of open-source datasets availability



The need for molecular representations



Representation

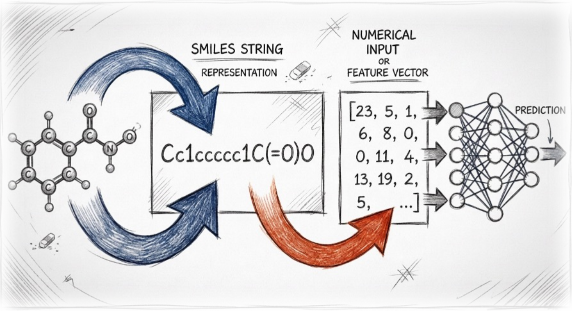


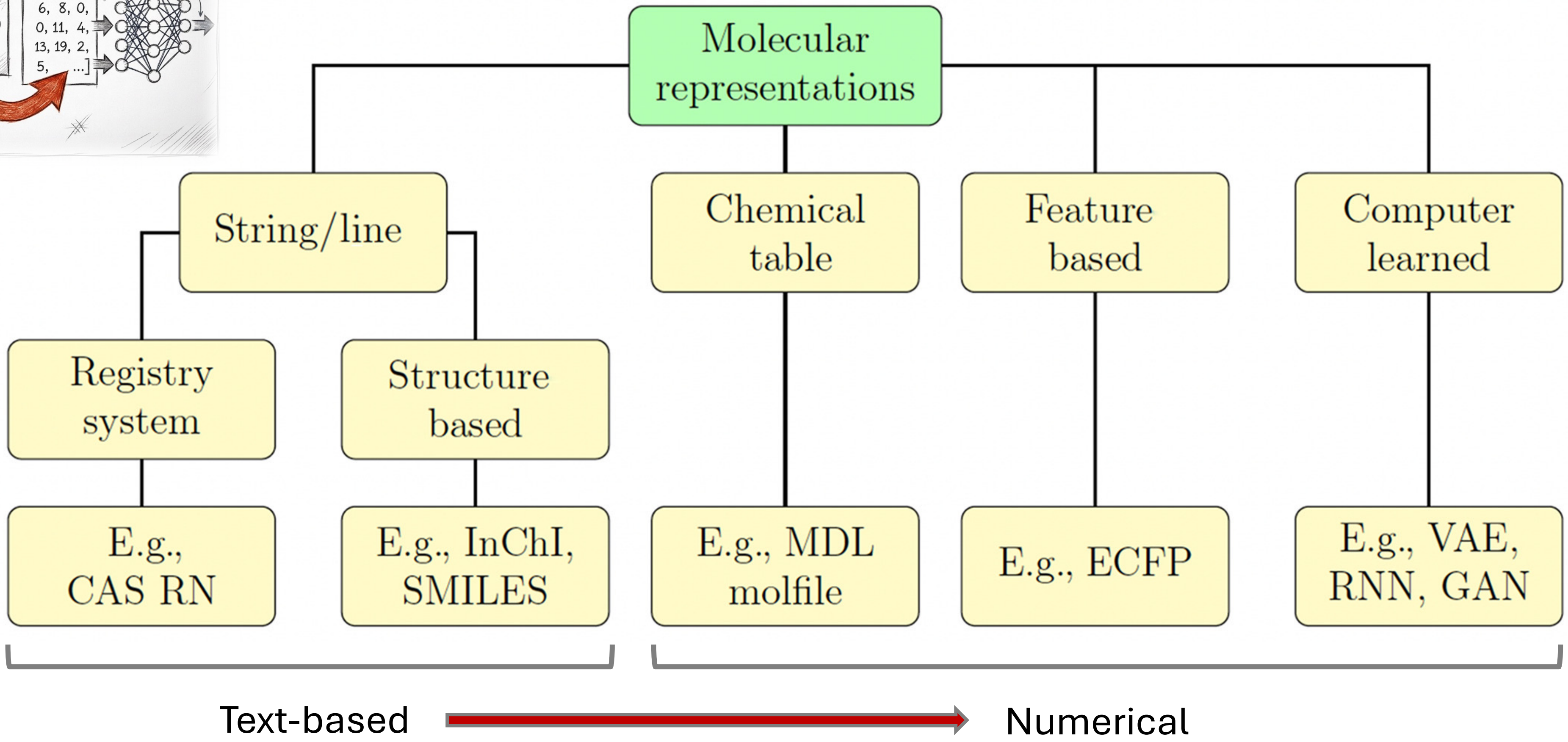
Optimize
Function/Property

Understand
Structure Activity
Relationship

information the model has
on top of the data – choices
to make – never neutral!

Type of molecular representations





Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. *WIREs Comput Mol Sci* **2022**, 12 (5).

The Evolution of Data-driven Chemistry

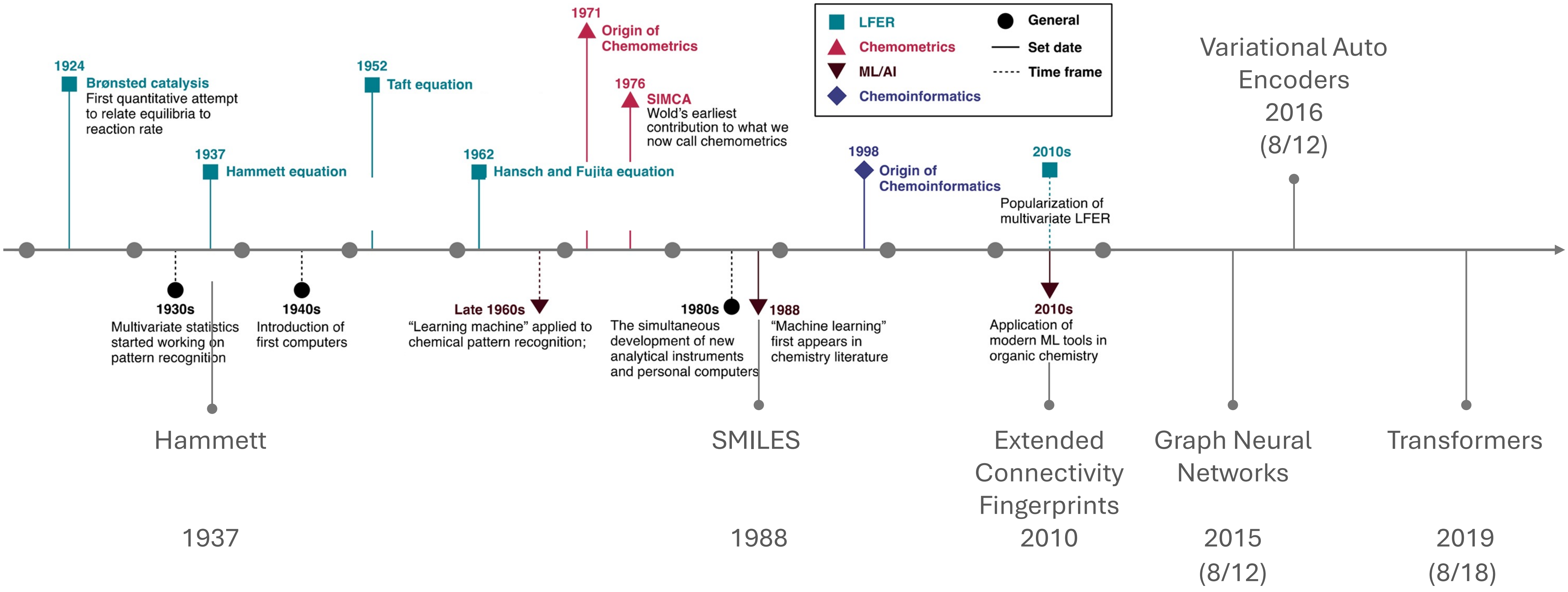
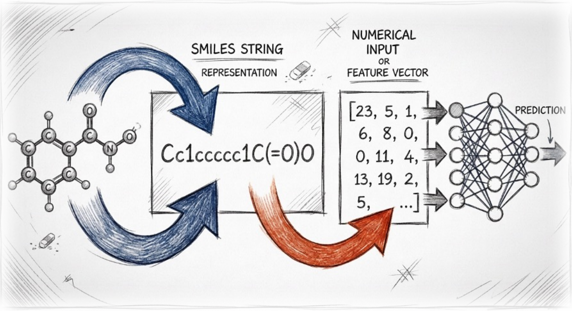


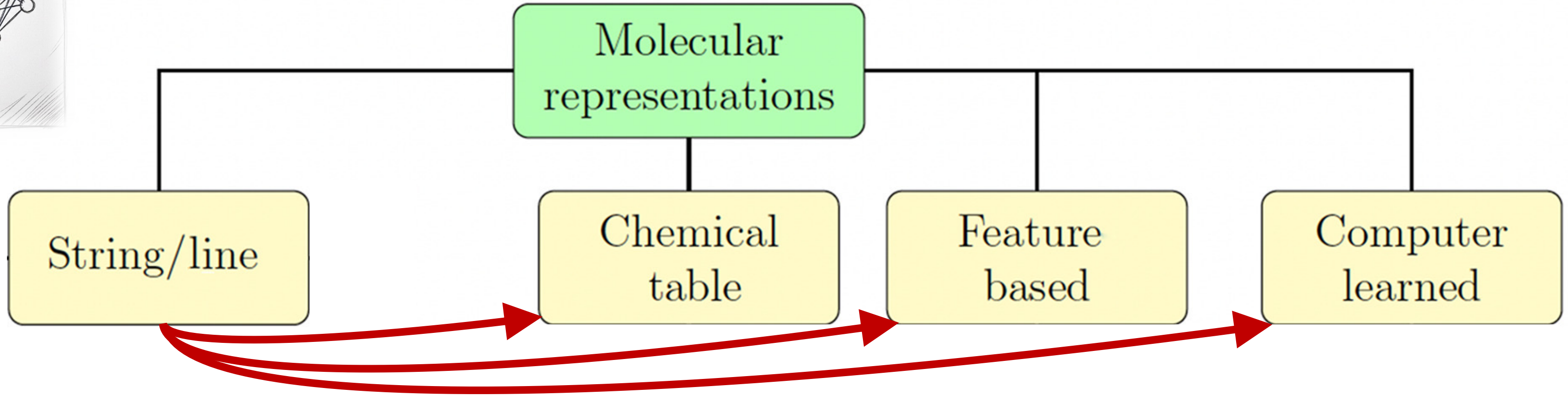
Figure adapted from Williams, W. L.; Zeng, L.; Gensch, T.; Sigman, M. S.; Doyle, A. G.; Anslyn, E. V. *ACS Cent. Sci.* **2021**, 7 (10), 1622–1637

Plan

- Evolution of “data-driven” chemistry
- **Overview of existing representations**
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- Data visualization

String based representations are usually the starting point



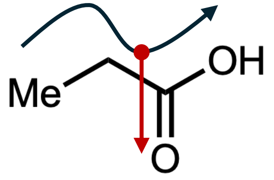


Small size
Easy to use
Potential missing Information

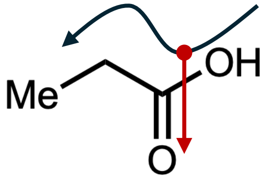
Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. *WIREs Comput Mol Sci* **2022**, 12 (5).

String line representations for small molecules: SMILES

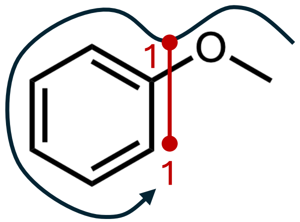
SMILES — Simplified Molecular-Input Line-Entry System



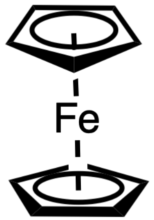
CCC(=O)O



OC(=O)CC



COc1ccccc1



C1=C[CH]C=C1.C1=C[CH]C=C1.[Fe]

SMILES Manipulation

OpenBabel

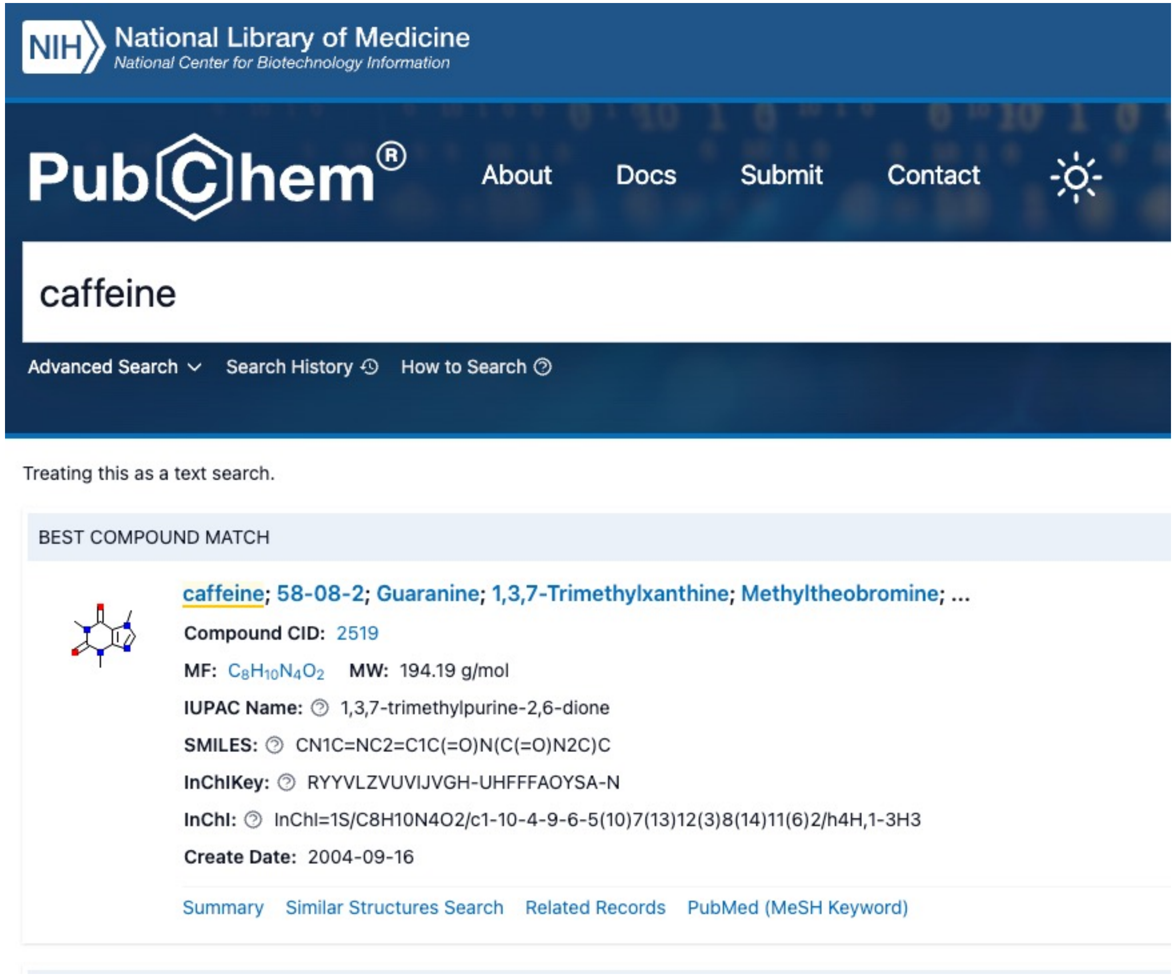
RDKit

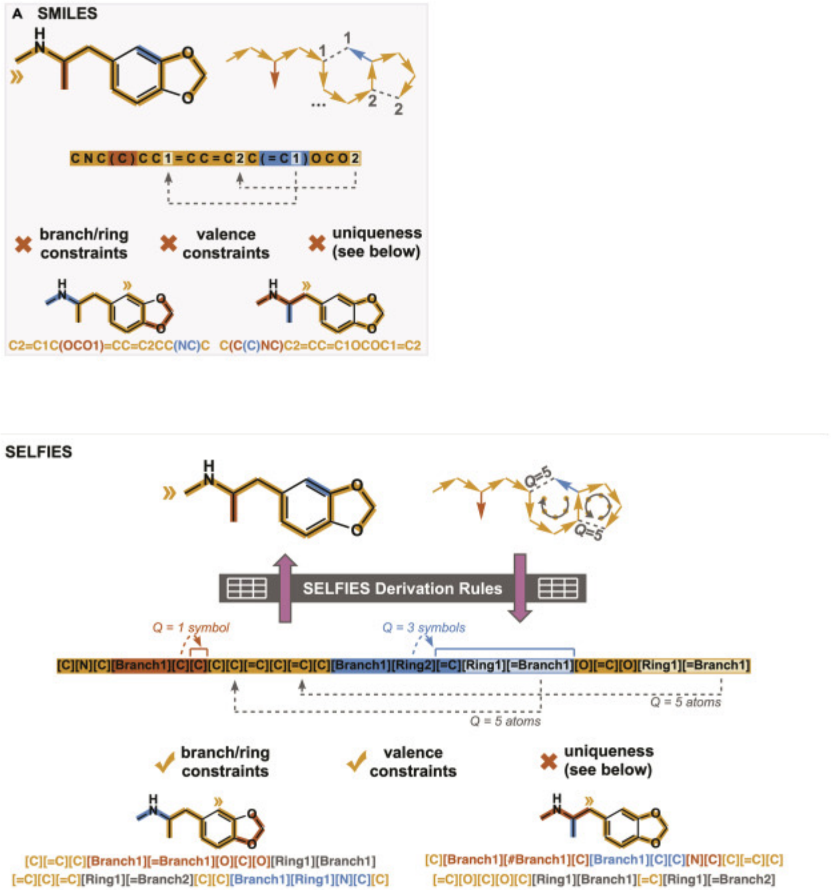
Open-Source Cheminformatics and Machine Learning

Weininger, D. *J. Chem. Inf. Comput. Sci.* **1988**, 28 (1), 31–36

https://openbabel.org/index.html
https://www.rdkit.org/

String base representations for small molecules: beyond SMILES

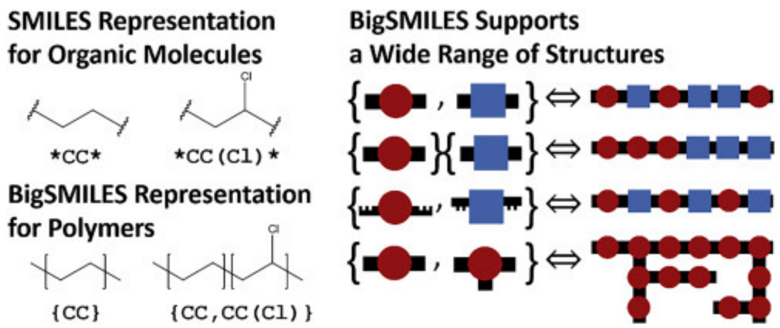




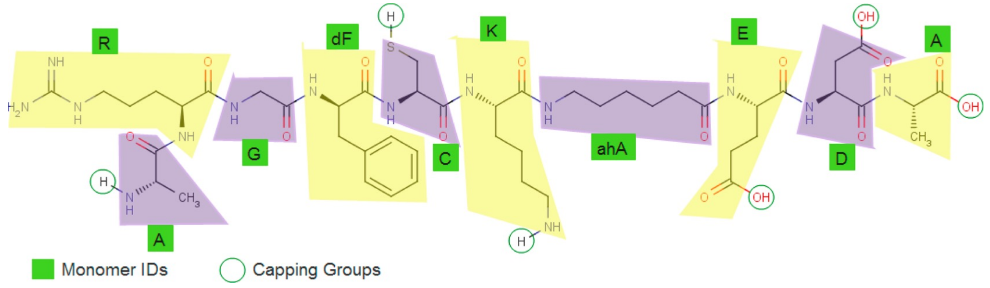
Krenn, M.; Häse, F.; Nigam, A.; Friederich, P.; Aspuru-Guzik, A. *Mach. Learn.: Sci. Technol.* **2020**, 1 (4), 045024.
https://pubchem.ncbi.nlm.nih.gov/

String base representations beyond small molecules

Polymers (BigSMILES)



Proteins (HELM)



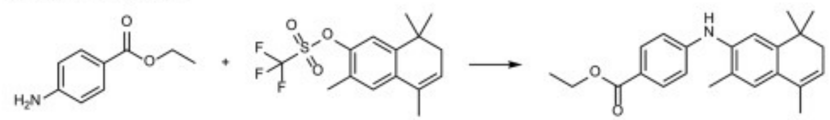
A.R.G.[dF].C.K.[ahA].E.D.A

Materials

SLICES (Simplified Line-Input Crystal-Encoding System)

Wyckoff-position strings

Reactions

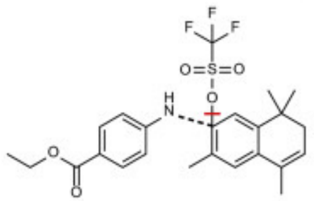


CCOC(=O)c1ccc(N)cc1.CC1=CCC(C)(C)c2cc(OS(=O)(=O)C(F)(F)F)c(C)cc21
>>CCOC(=O)c1ccc(Nc2cc3c(cc2C)C(C)=CCC3(C)C)cc1

Reactants: CCOC(=O)c1ccc(N)cc1.CC1=CCC(C)(C)c2cc(OS(=O)(=O)C(F)(F)F)c(C)cc21

Product: CCOC(=O)c1ccc(Nc2cc3c(cc2C)C(C)=CCC3(C)C)cc1

Condensed graph of reaction (requires atom-mapping):



FC(F)(F)S(=O)(=O)O[->.]c1(cc2C(C)(C)CC=C(c2cc1C)C)[.>-]Nc3ccc(cc3)C(OCC)=O

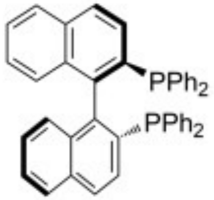
broken bond *formed bond*

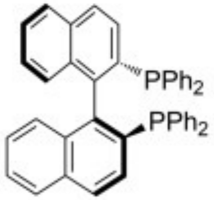
Krenn, M. et al. *Patterns* **2022**, 3 (10), 100588.

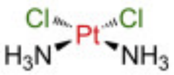
Ongoing issues with string based representations

Stereoisomers

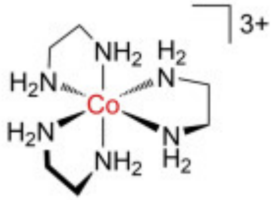

[7]helicene

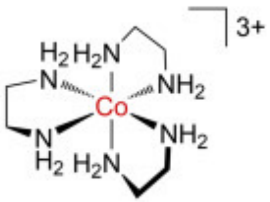

(*R*)-BINAP


(*S*)-BINAP

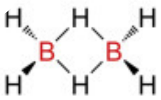

cisplatin

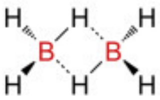

transplatin

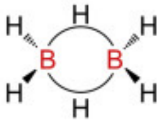

Λ-isomer

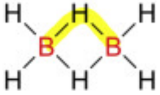

Δ-isomer

Hapticity

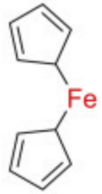

1


2

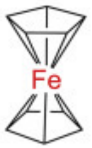

3


4


5


6


7

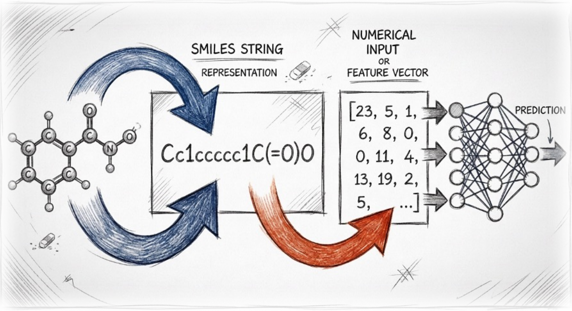

8

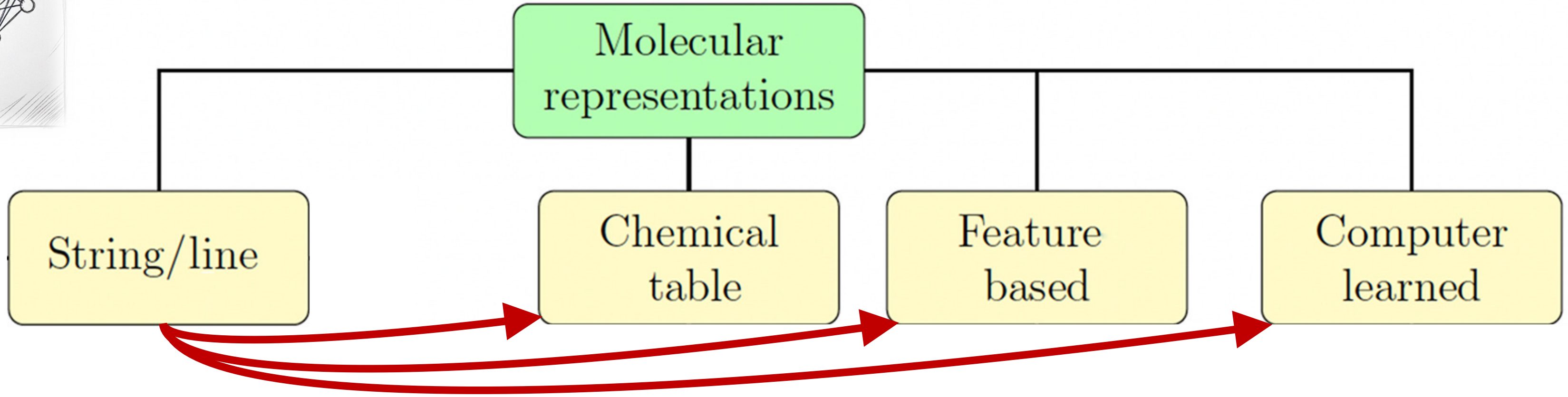
Be aware of representation limitations!

Which information is/is not captured in my representation?

Krenn, M. et al. *Patterns* **2022**, 3 (10), 100588.

From String Based Representation to Numeric Representations





Numerisation/Vectorization of
the Information

In the literature descriptors or features are both used for “numerical representations”

Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. *WIREs Comput Mol Sci* **2022**, 12 (5).

Descriptors Survey

I. Small Molecule Descriptors

II. Reaction Descriptors

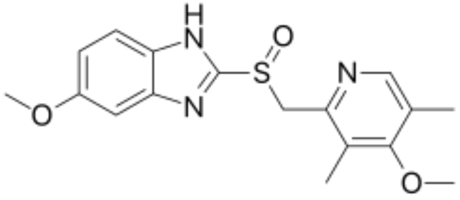
III. Materials Descriptors

Numeric representations: Small Molecules

Lipinski's rule of 5 for drug likeness:

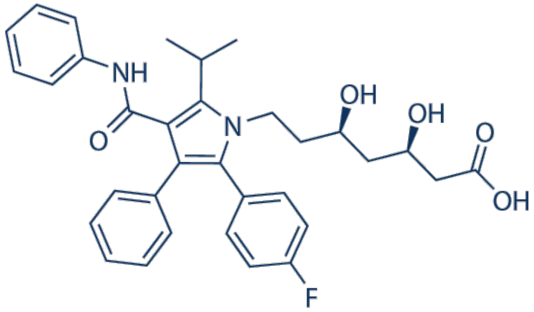
- No more than 5 hydrogen bond donors
- No more than 10 hydrogen bonds acceptors
- Molecular mass less than 500 daltons
- Partition coefficient water-octanol (ClogP) < 5

Omeprazole



- 1
- 5
- 345 g/mol
- 2.3 – 2.6

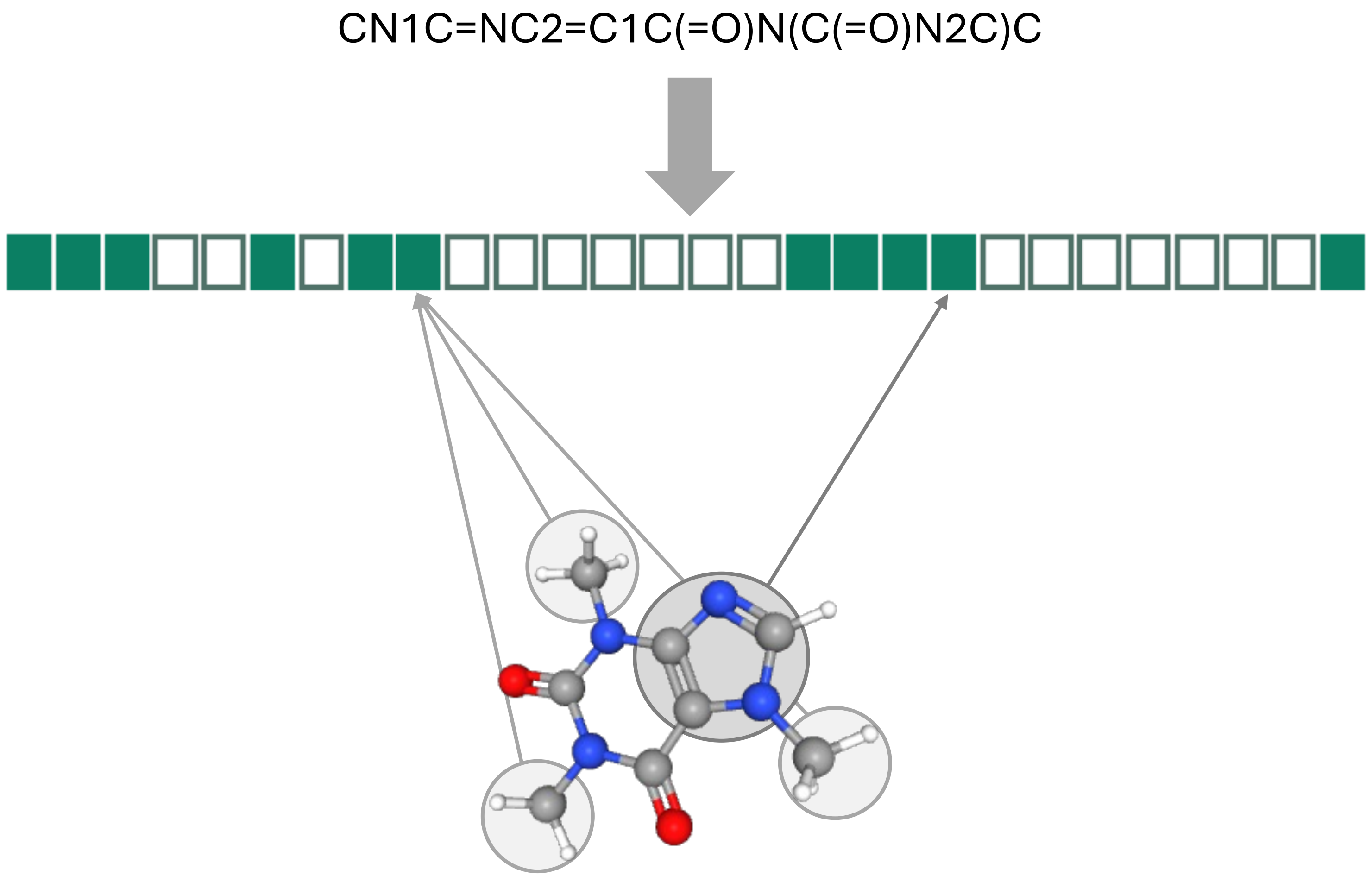
Atorvastatin



- 4
- 56
- 559 g/mol
- 2.2 – 2.56

Numeric representations: Small Molecules

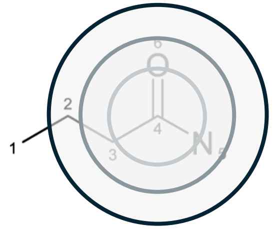
Molecular Fingerprints Developed for Quantitative Structure Activity Relationships (QSAR):



Advantages:
- Very fast to compute
- Tanimoto/Jaccard similarity

Information encoded:
- 2D structure/connectivity
- Stereochemistry
- No 3D information
- No conformers

Extended Connectivity Fingerprints (ECFPs):



Radius 0

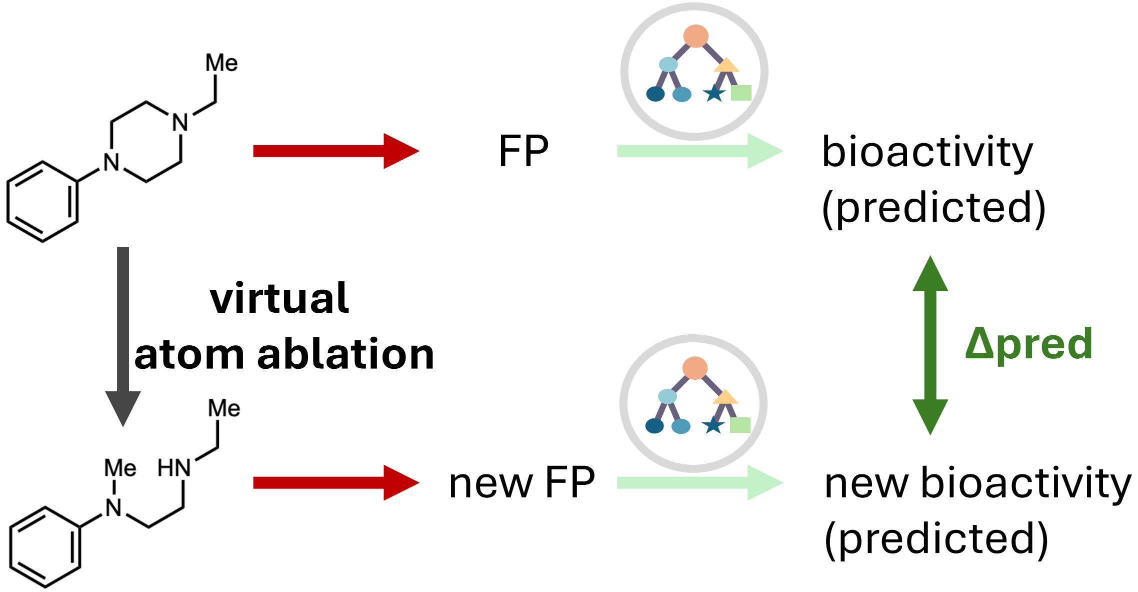
Radius 1

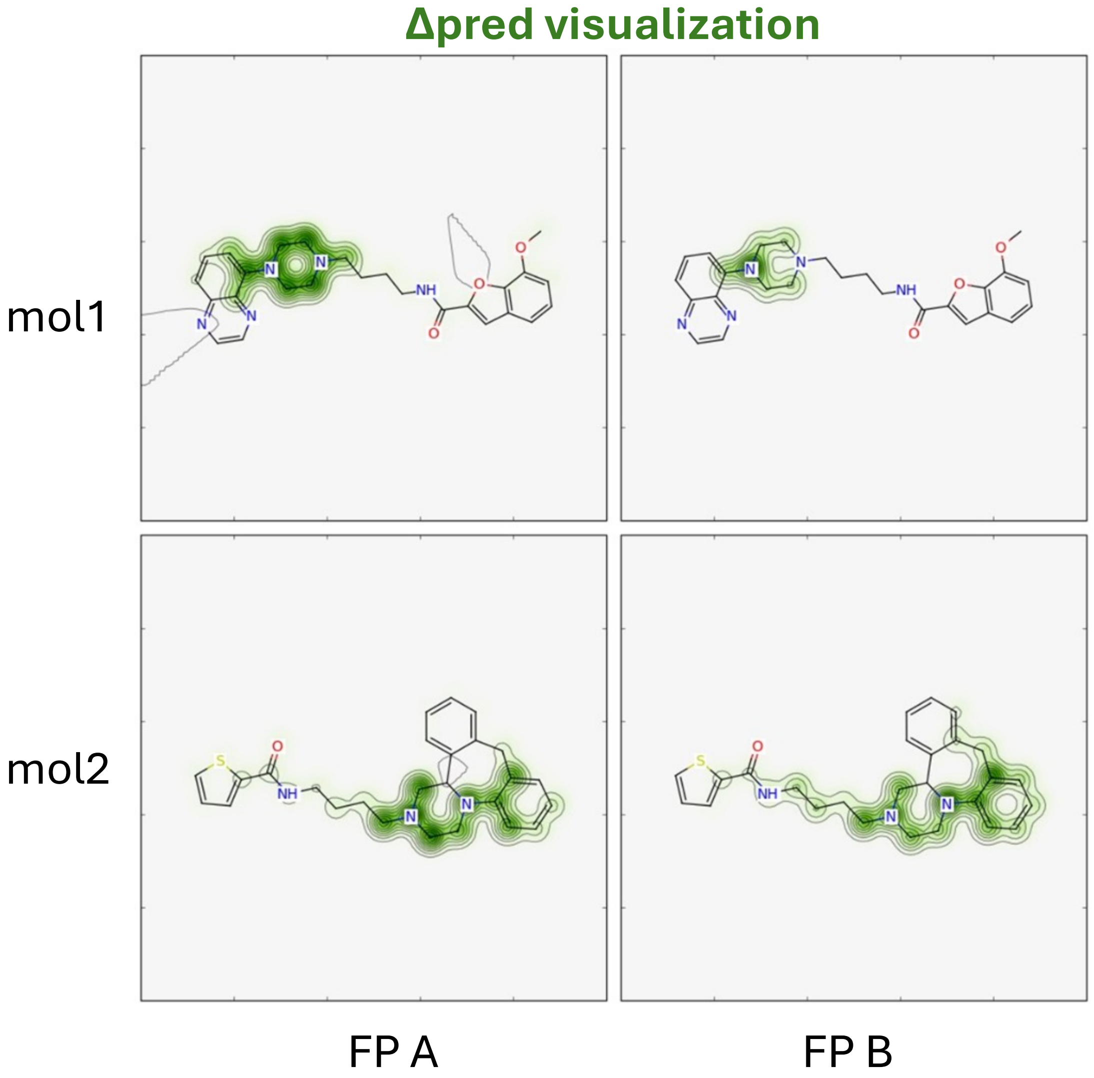
Radius 2

Rogers, D.; Hahn, M. *J. Chem. Inf. Model.* **2010**, 50 (5), 742–754

More on FPs generation: https://greglandrum.github.io/rdkit-blog/posts/2023-01-18-fingerprint-generator-tutorial.html

Application ECFPs for QSAR





Riniker, S.; Landrum, G. A. *J Cheminform* **2013**, 5 (1).

Recap on Molecular Fingerprints

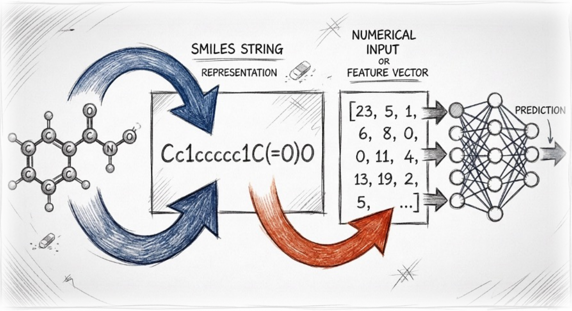
When to use:

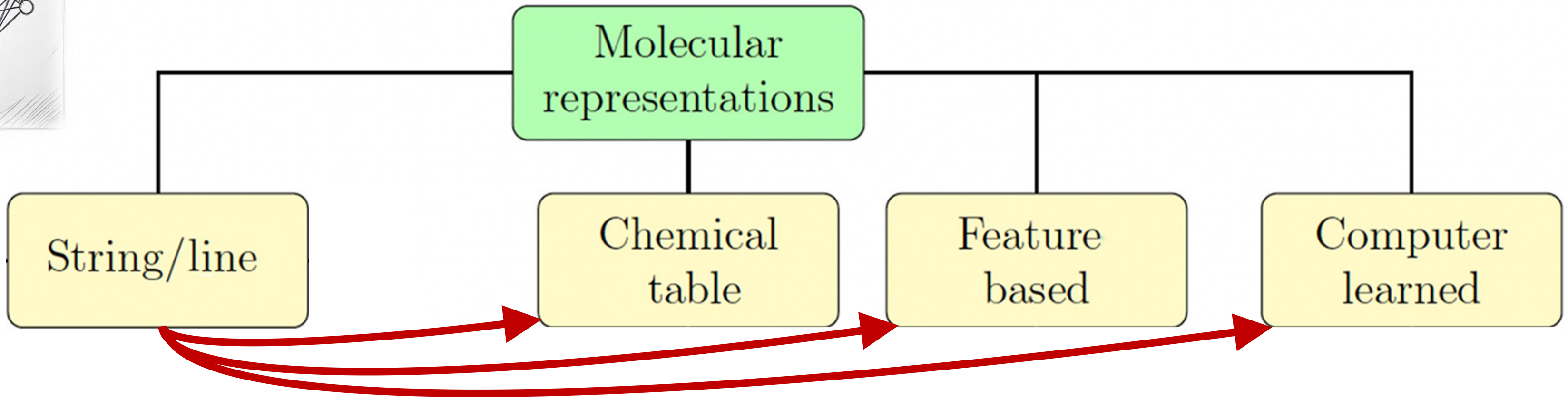
- Good for a fast baseline small molecules activity
- Efficient with relatively small datasets (100 – 10k molecules)

When not to use:

- Properties/Activities depend on 3-D shape or conformers
- Very small datasets
- Non-organic molecules/other chemical systems

From String Based Representation to Numeric Representations



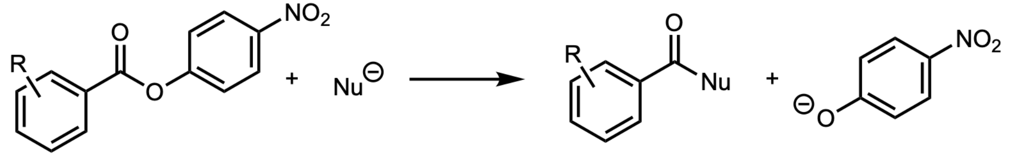


Numerisation/Vectorization of
the Information

Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. *WIREs Comput Mol Sci* **2022**, 12 (5).

Feature Based Representations: Experimental Descriptors

Hammett descriptors:



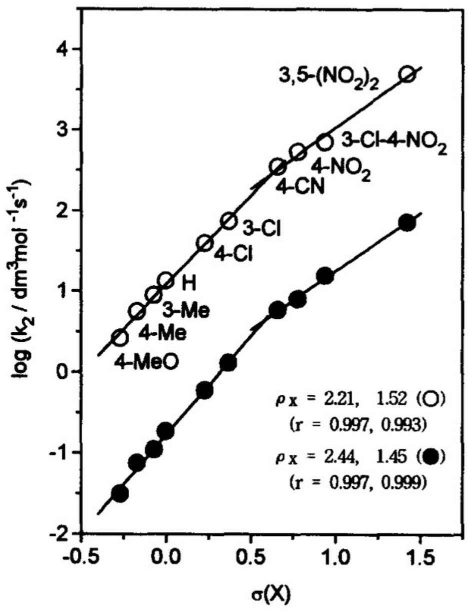


Figure 1. Hammett plots for the reactions of *p*-nitrophenyl X-substituted benzoates with OH^-(○) and *p*-chlorophenoxide(●) in H_2O containing 20 mole % DMSO at 25.0±0.1℃.

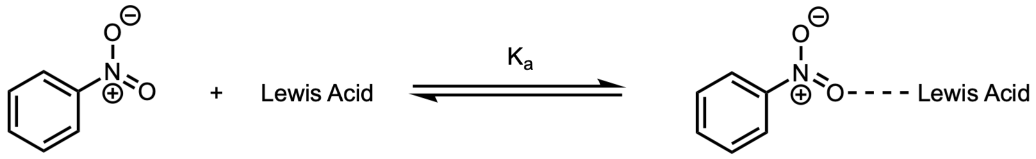
Advantages:
- Mechanism interpretability
- Condensed information
- Small dataset interpretability

Limitations:
- Specific electronic effects
- Availability

Um, I.-H.; Chung, E.-K.; Kwon, D.-S. *Tetrahedron Letters* **1997**, 38 (27), 4787–4790.

Feature Based Representations: Experimental Descriptors

Predicting Lewis Acid Base Affinity with Experimental Descriptors



NMR Acceptor Number (AN)
Water Exchange Rate (WERC)
pK_h
Electrostatic Acceptor Number (E_a)
Covalent Acceptor Number (C_a)
Infrared Xanthone Complexes (IrX)
Infrared TPPO Complexes (IrTPPO)
Infrared EthylAcetate Complexes (IrAcOEt)
Oxidation Peak (Ox)
Complexing Power (CP)

Advantages:
- Mechanism interpretability
- Condensed information
- Small dataset interpretability

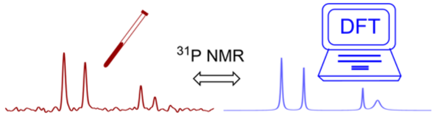
Limitations:
- Specific electronic effects
- Availability of experimental descriptors

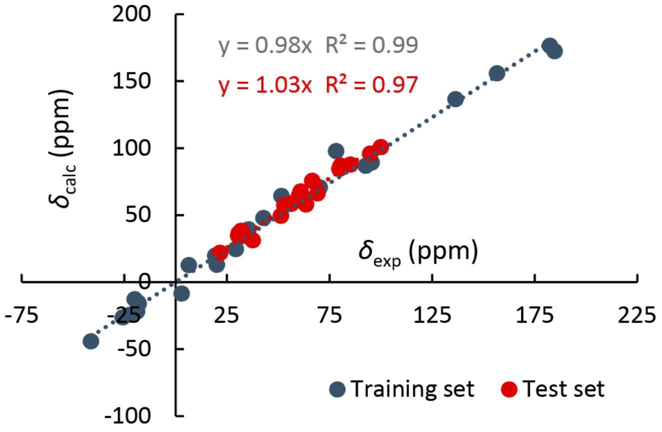
Today:
- Computed descriptors

J. Schleinitz, https://theses.hal.science/tel-05018662

Feature Based Representations: Computed Descriptors

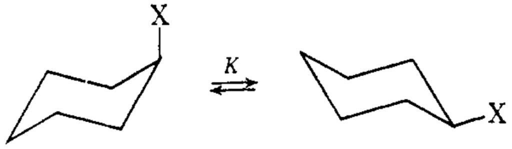
Phosphine Featurization by DFT:





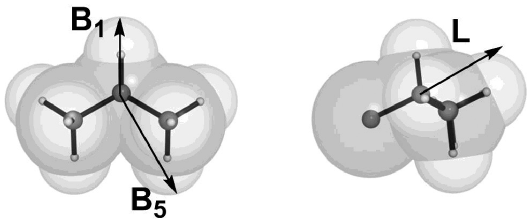
Geometries optimized at the B3LYP/BS1 level of theory. ^{31}P NMR shielding constants were evaluated with the GIAO method at the B3LYP/BS2 level of theory.

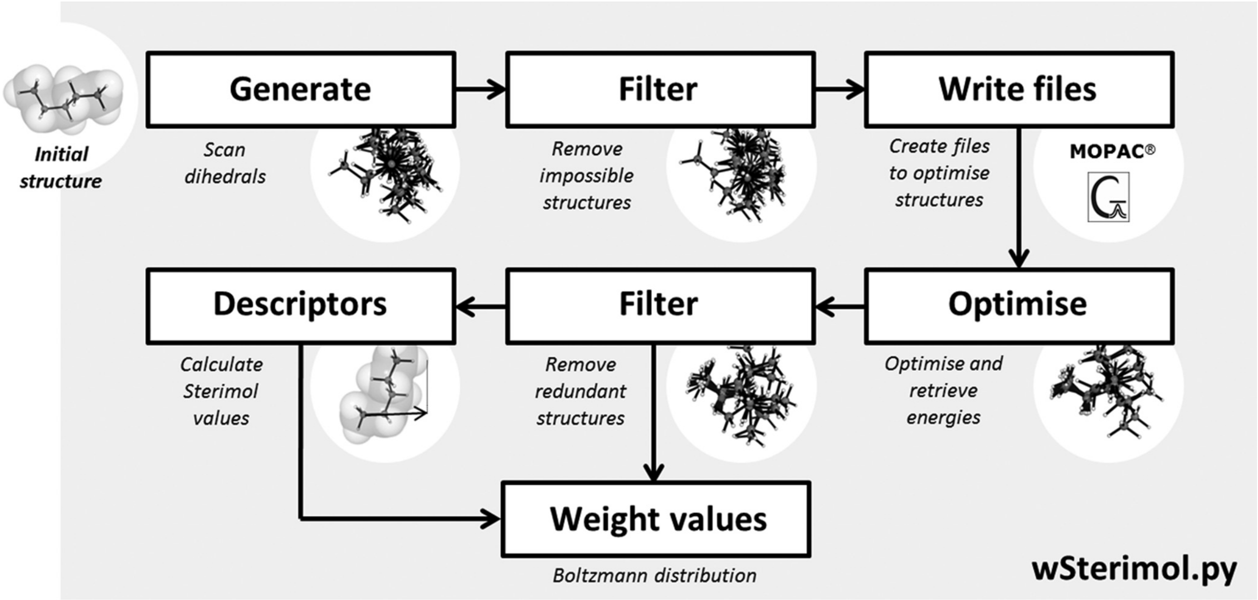
Steric Effects:



$$A = -\Delta F^{\circ} = RT \ln K$$

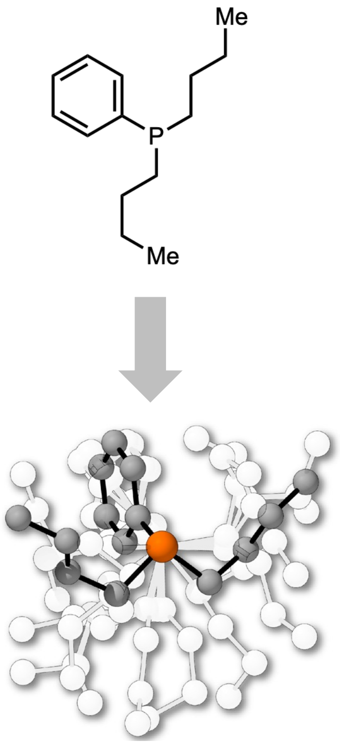
Sterimol Descriptors:



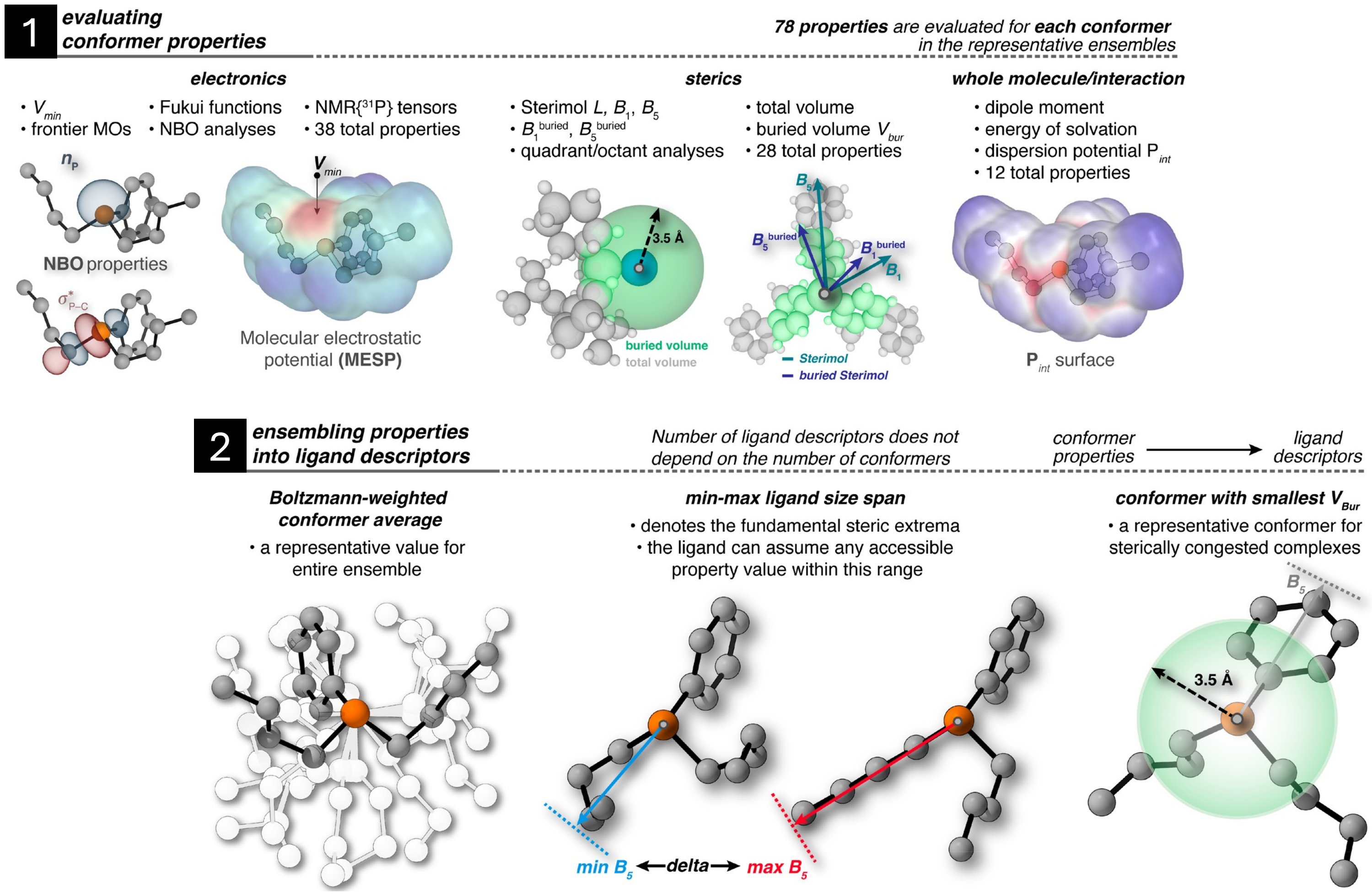


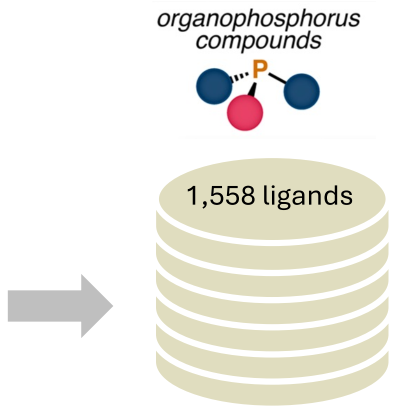
Payard, P.-A.; Perego, L. A.; Grimaud, L.; Ciofini, I. *Organometallics* **2020**, 39 (17), 3121–3130.
Jensen, F. R.; Bushweller, C. H.; Beck, B. H. *J. Am. Chem. Soc.* **1969**, 91(2), 344–351
Brethomé, A. V.; Fletcher, S. P.; Paton, R. S. *ACS Catal.* **2019**, 9 (3), 2313–2323.

Feature Based Representations: Descriptors Libraries



Conformer ensemble generation

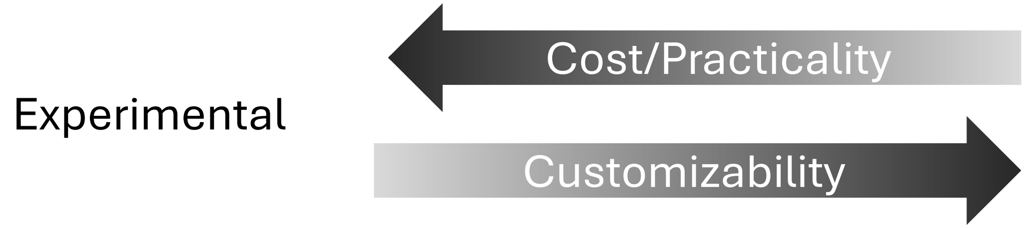


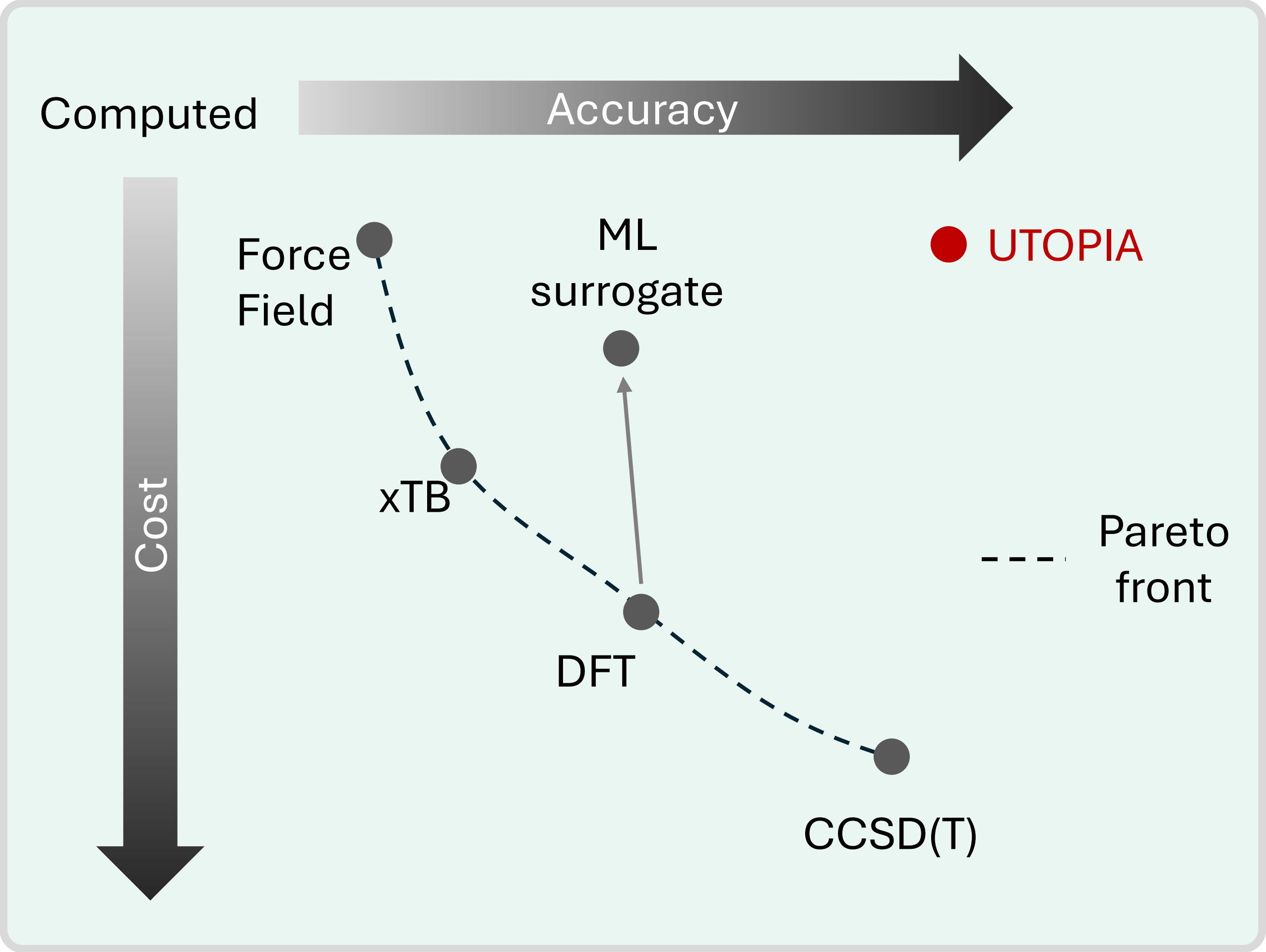


Descriptor Database

Gensch, T.; [...] Sigman, M. S.; Aspuru-Guzik, A. *J. Am. Chem. Soc*. **2022**, 144 (3), 1205–1217.
https://descriptor-libraries.molssi.org/

Feature Based Representations: Towards Predicted Descriptors





Examples of ML-Surrogates For Descriptor Generation

MLIPs for Geometry Optimization (Wednesday!)

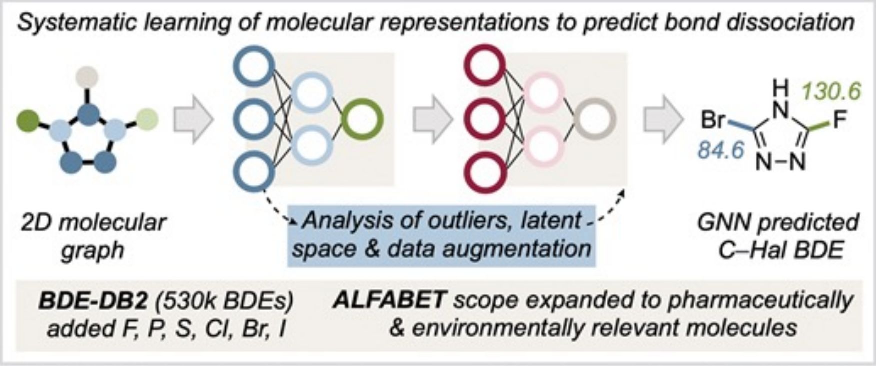
BDE predictions:
- BondNET charged species
- Alfabet (homolytic neutral molecules)

Solubility:

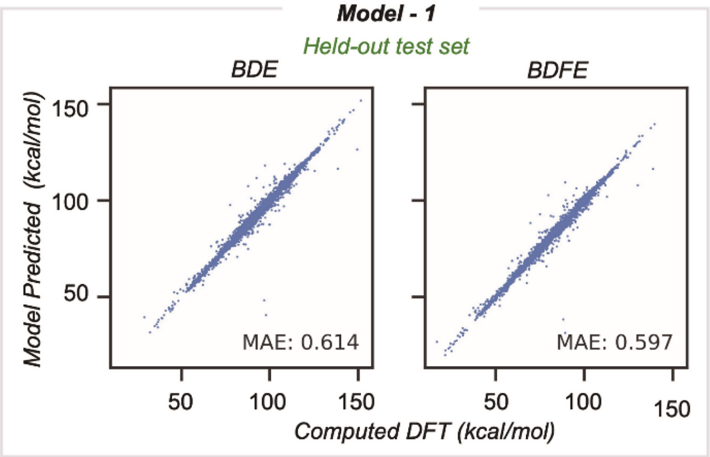
pKa predictions:
MolGpKa (https://xundrug.cn/molgpka)

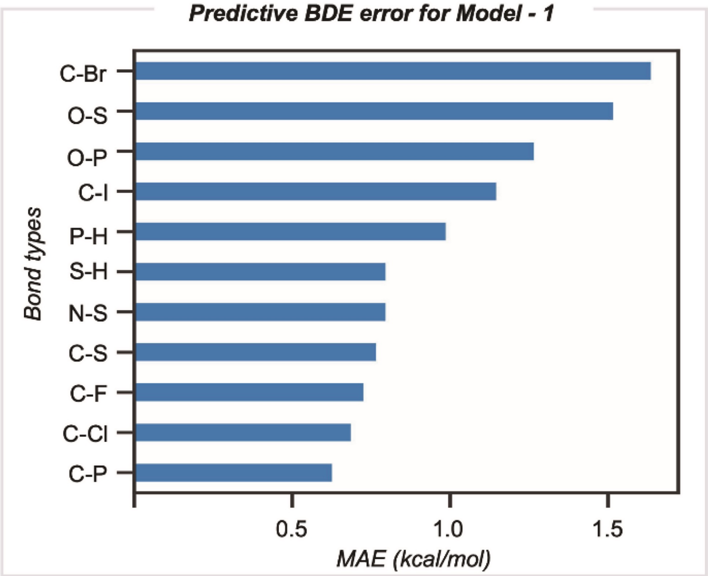
Prediction of BDEs using ML:

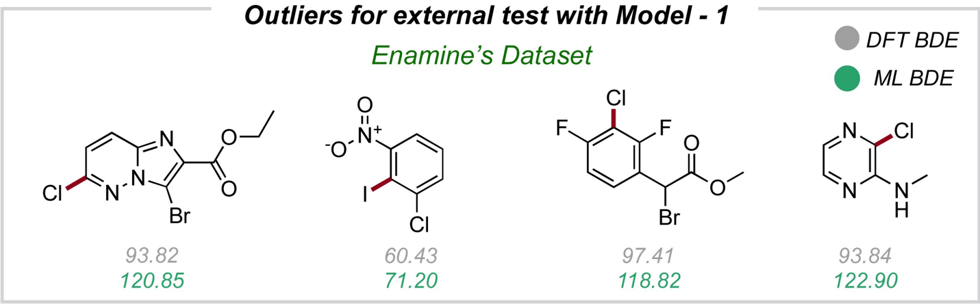
ML pipeline:



Validation and Domain of Applicability





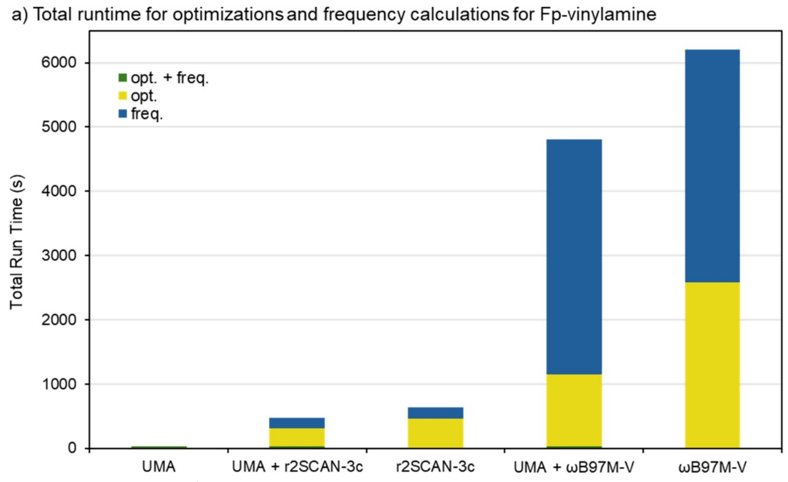


Always evaluate validity
Recompute some data points with the high accuracy method for important points!

S. V., S. S.; Kim, Y.; Kim, S.; St. John, P. C.; Paton, R. S. *Digital Discovery* **2023**, 2 (6), 1900–1910.

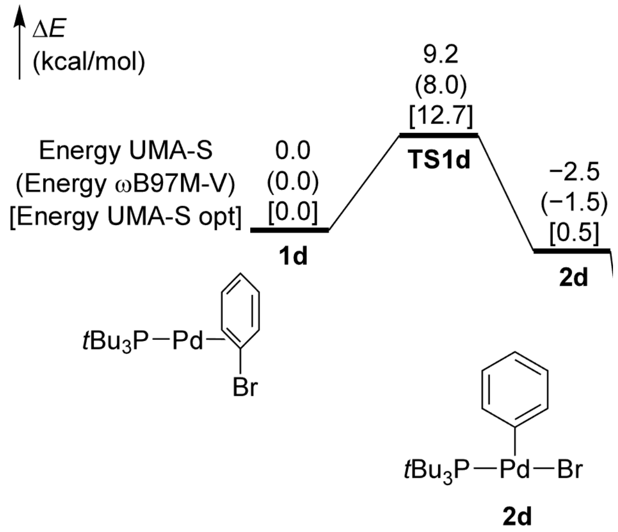
Expert Crafted Descriptors: Using ML models

Machine Learning Interatomic Potentials as DFT surrogate (Structure Optimization/Energy Calculation):

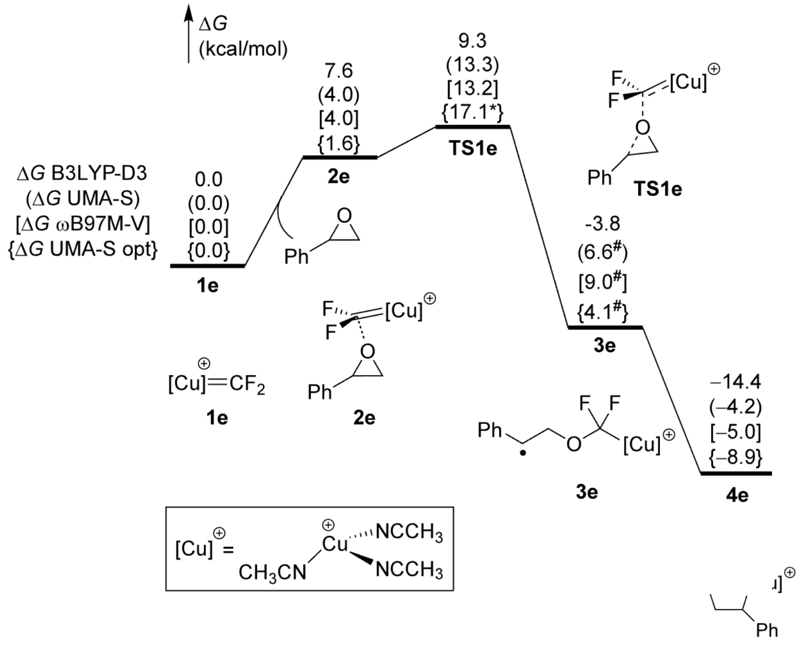


>99% reduction compute

Closed shell

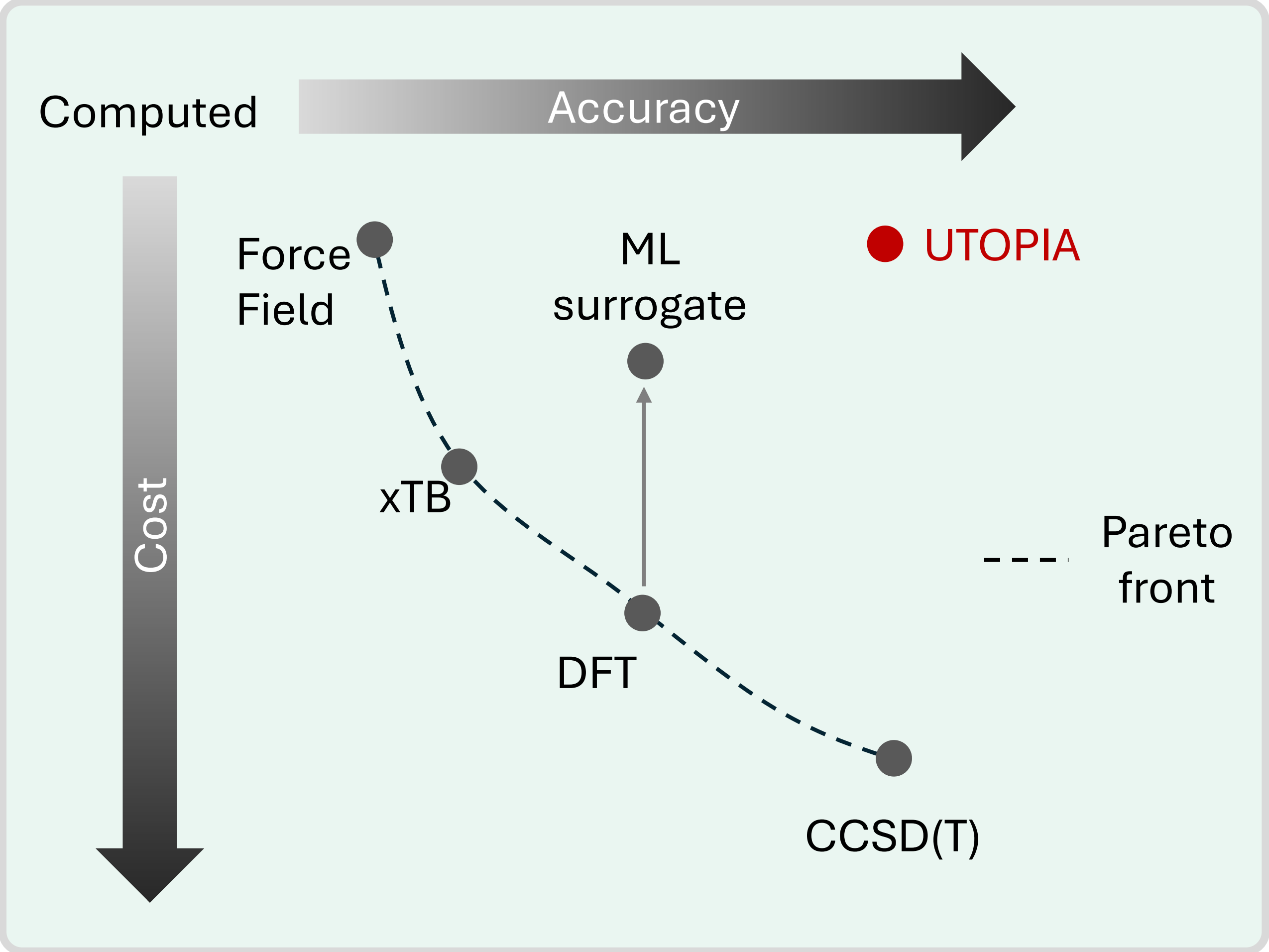


Open shell



Quirion, K. P.; Kong, W.-Y.; Stanley, B.; Joy, J.; Ess, D. H. *ACS Catal.* **2026**. 16 (9): 8053–8066.

Expert Crafted Descriptors: Using ML models



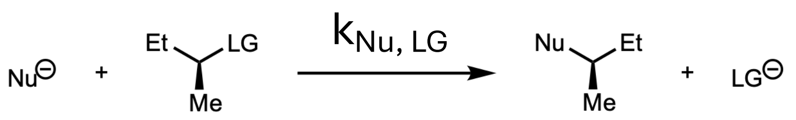
Ab-initio or semi-empirical computational tools have identifiable flaws/literature experience.

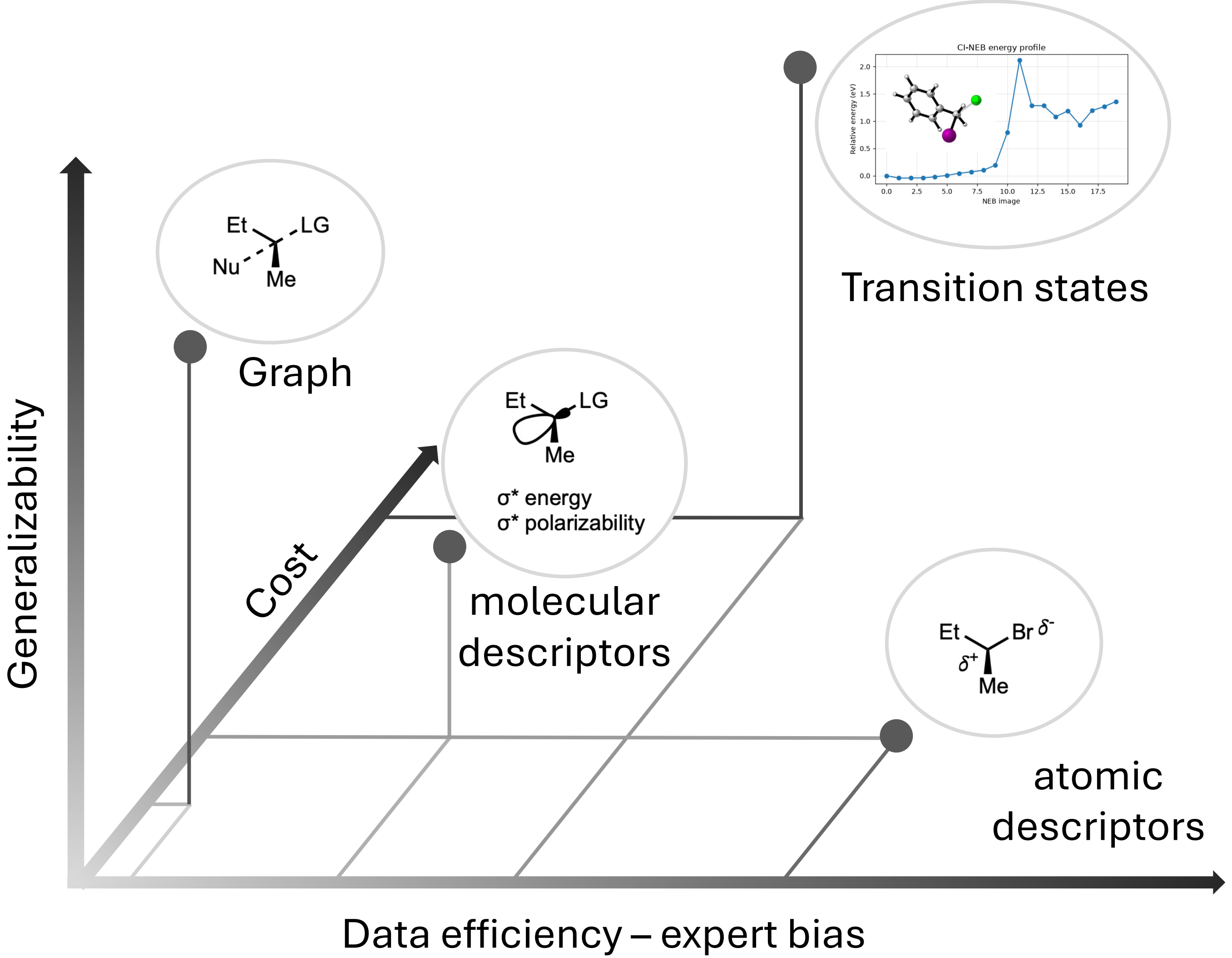
ML surrogates are less predictable: Good usage practice:

- Benchmark few points vs DFT/reference method
- Understand/evaluate domain of applicability

Expert-Crafter Descriptors: Precision/Generalizability Trade-off

Descriptors for modeling rates of SN_2?

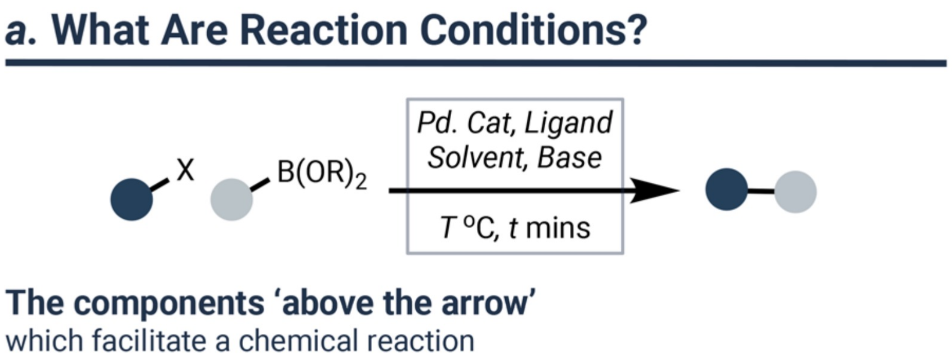




Multi-parameter/Combinatorial Chemical Systems

Example of Organic Reaction Conditions

Concatenate descriptors of reactant, conditions and product



Encode the reaction as a whole

Ball, M.; Horvath, D.; Kogej, T.; Kabeshov, M.; Varnek, A. Predicting Reaction Conditions: A Data-Driven Perspective. *Chem. Sci.* **2025**, 16 (38), 17523–17541.

Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

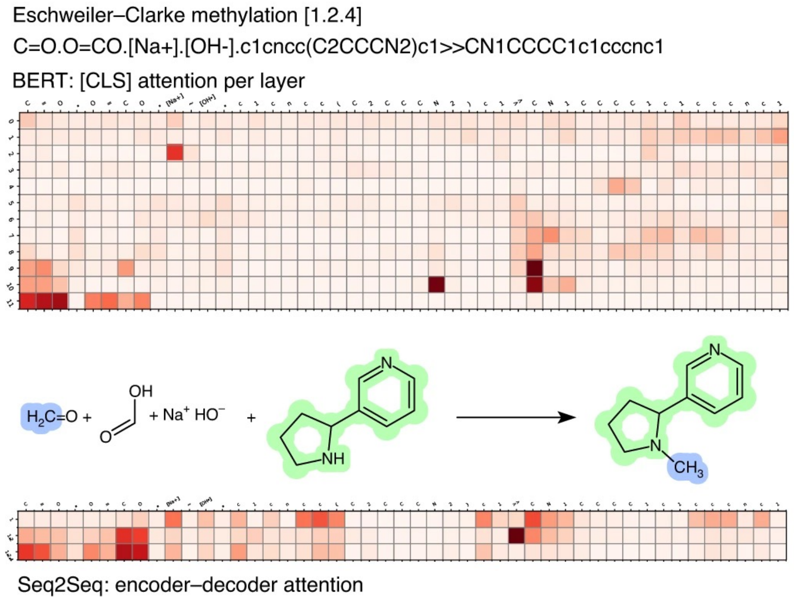
Encode the reaction as a whole:

Reaction SMILES:

BrC1CCCCC1=O>C[O-].[Na+]>COC(=O)C1CCCC1

Differential Reaction Fingerprints

Learned Encoding (Transformers)



Advantages:

- Universal
- Interpretable (DRFP)

Disadvantages:

- Coarse grained information
- Requires large datasets

Schneider, N.; Lowe, D. M.; Sayle, R. A.; Landrum, G. A. *J. Chem. Inf. Model.* **2015**, 55 (1), 39–53. / Probst, D.; Schwaller, P.; Reymond, J.-L. *Digital Discovery* **2022**, 1 (2), 91–97.
Schwaller, P.; Laino, T.; Gaudin, T.; Bolgar, P.; Hunter, C. A.; Bekas, C.; Lee, A. A. *ACS Cent. Sci.* **2019**, 5 (9), 1572–1583. /

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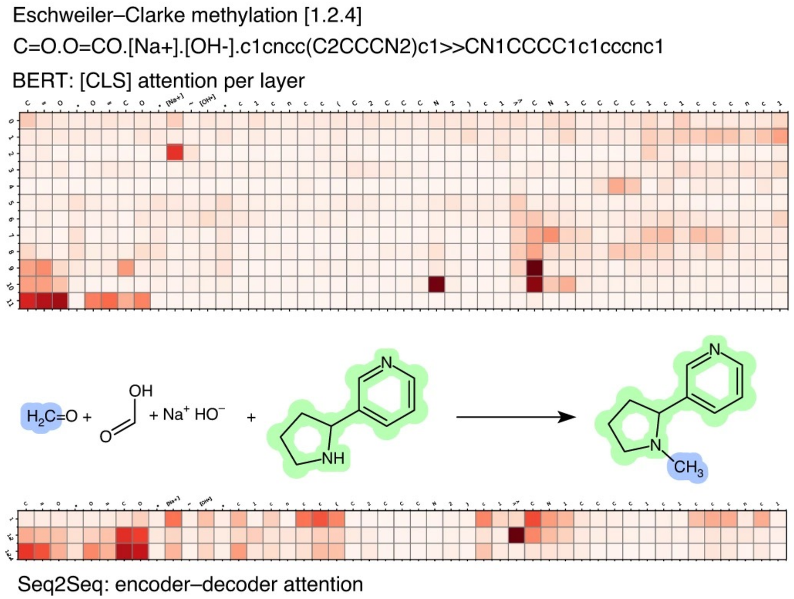

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Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

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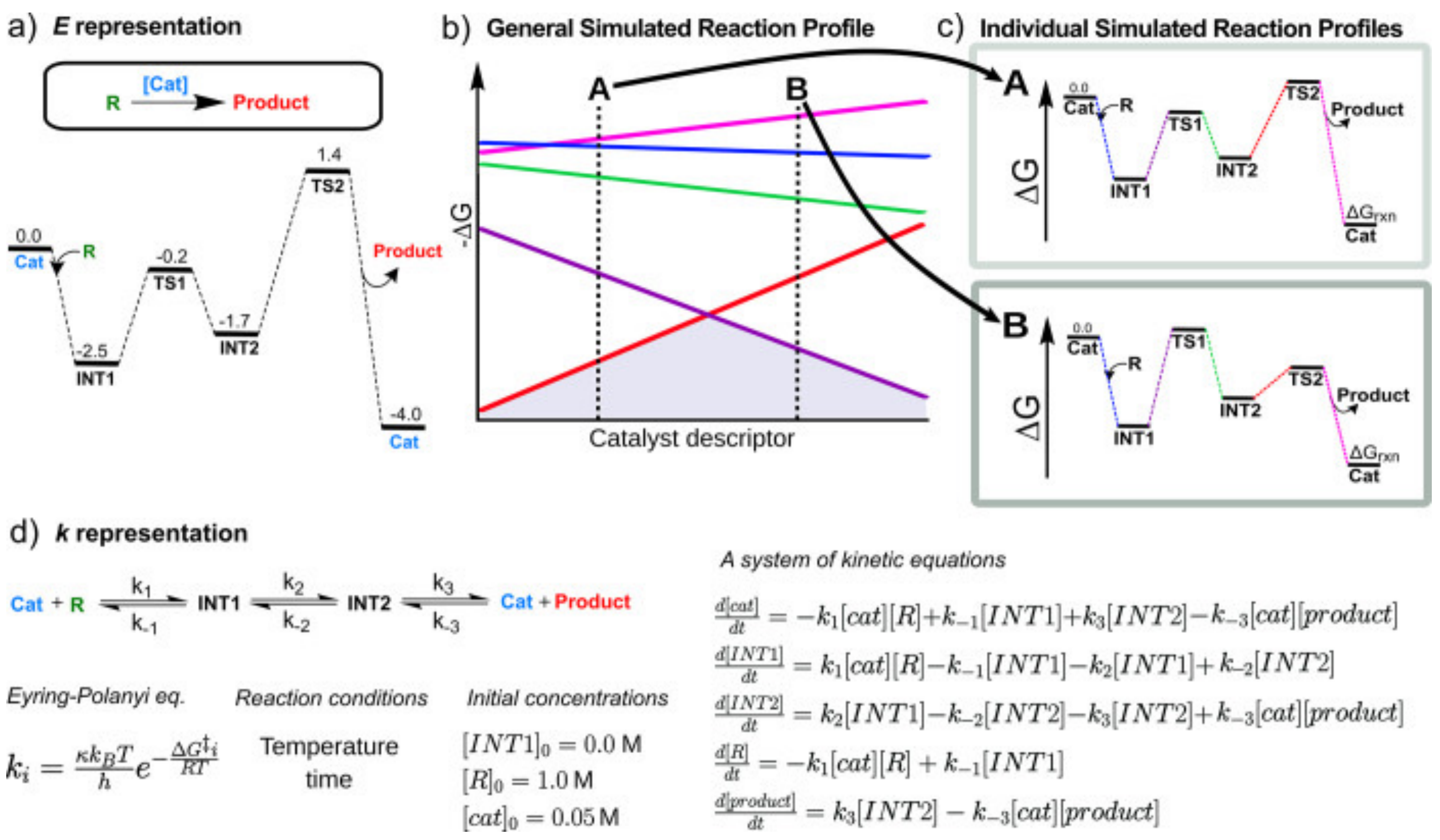


Advantages:

- Captures subtle reactivity trends
- Transferable between catalysts

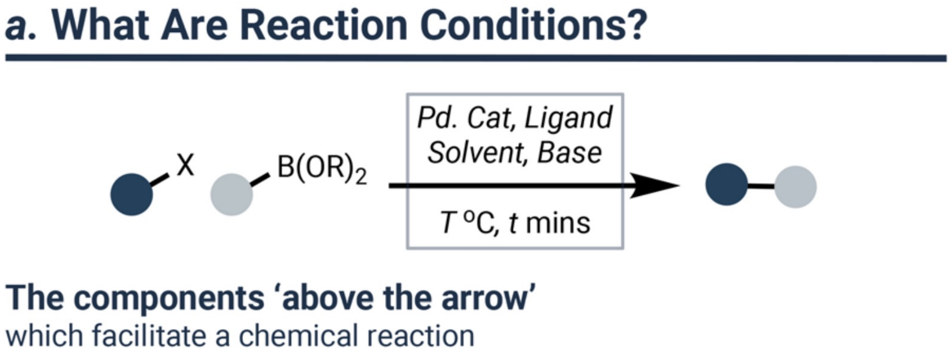
Disadvantages:

- Mechanism must be known/constant
- Computationally costly
- Only encodes catalyst

Worakul, T.; Laplaza, R.; Das, S.; Wodrich, M. D.; Corminboeuf, C. *ACS Catal*. **2024**, 14 (13), 9829–9839.

Example of Organic Reaction Conditions

Concatenate descriptors of reactant, conditions and product



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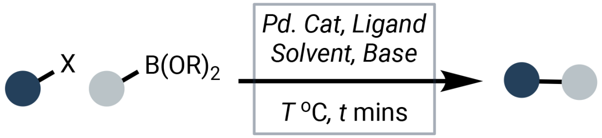
Ball, M.; Horvath, D.; Kogej, T.; Kabeshov, M.; Varnek, A. Predicting Reaction Conditions: A Data-Driven Perspective. *Chem. Sci.* **2025**, 16 (38), 17523–17541.

Multi-parameter/Combinatorial Chemical Systems: Organic Reactions

Integrating multiple variables?



The components 'above the arrow'

REAGENTS
CHEMICAL VARIABLES

Reacting species which **don't contribute a heavy atom** to the product.

- *Pd Cat.*
- *Ligand*
- *Solvent*
- *Base*

Categorical

PHYSICAL PARAMETERS
NON-CHEMICAL VARIABLES

Other non-chemical variables that influence a reaction.

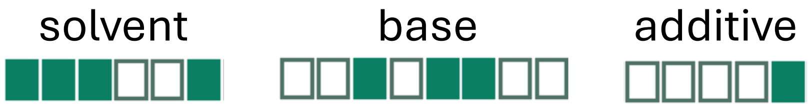
Here:
- *Temp.*
- *Time*

+ more…

Continuous

c. Mathematical Formulation

Concatenation of features:

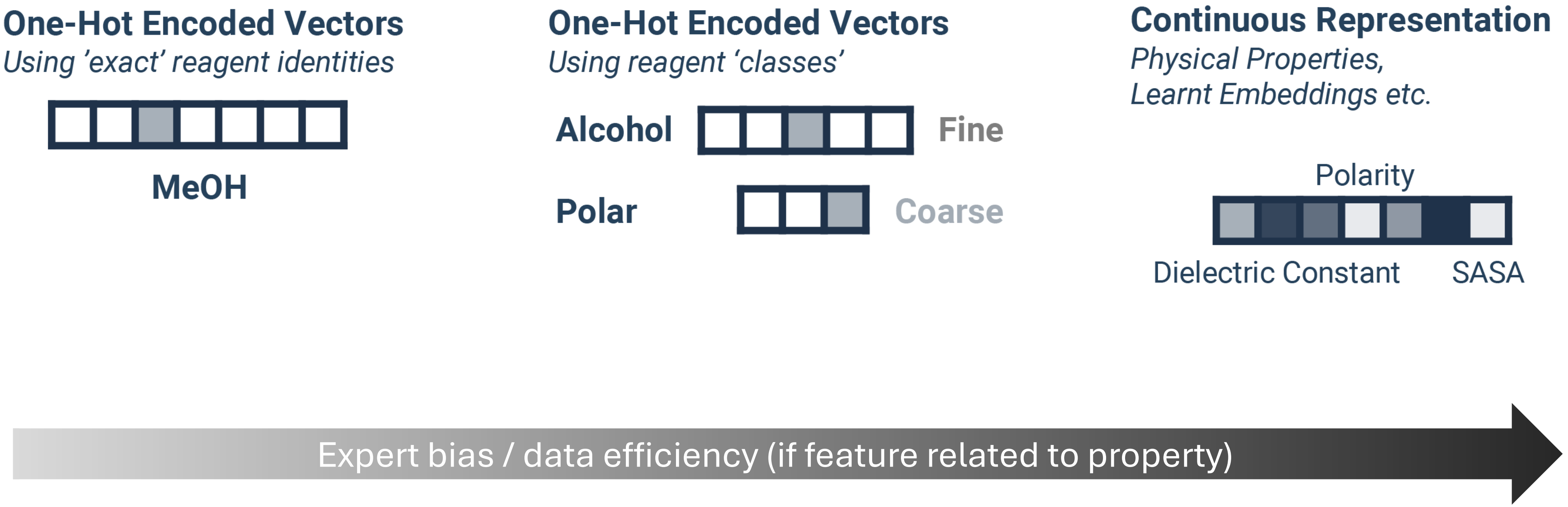


Choice of features: choosing the right amount of expert bias to use?

Ball, M.; Horvath, D.; Kogej, T.; Kabeshov, M.; Varnek, A. Predicting Reaction Conditions: A Data-Driven Perspective. *Chem. Sci.* **2025**, 16 (38), 17523–17541.

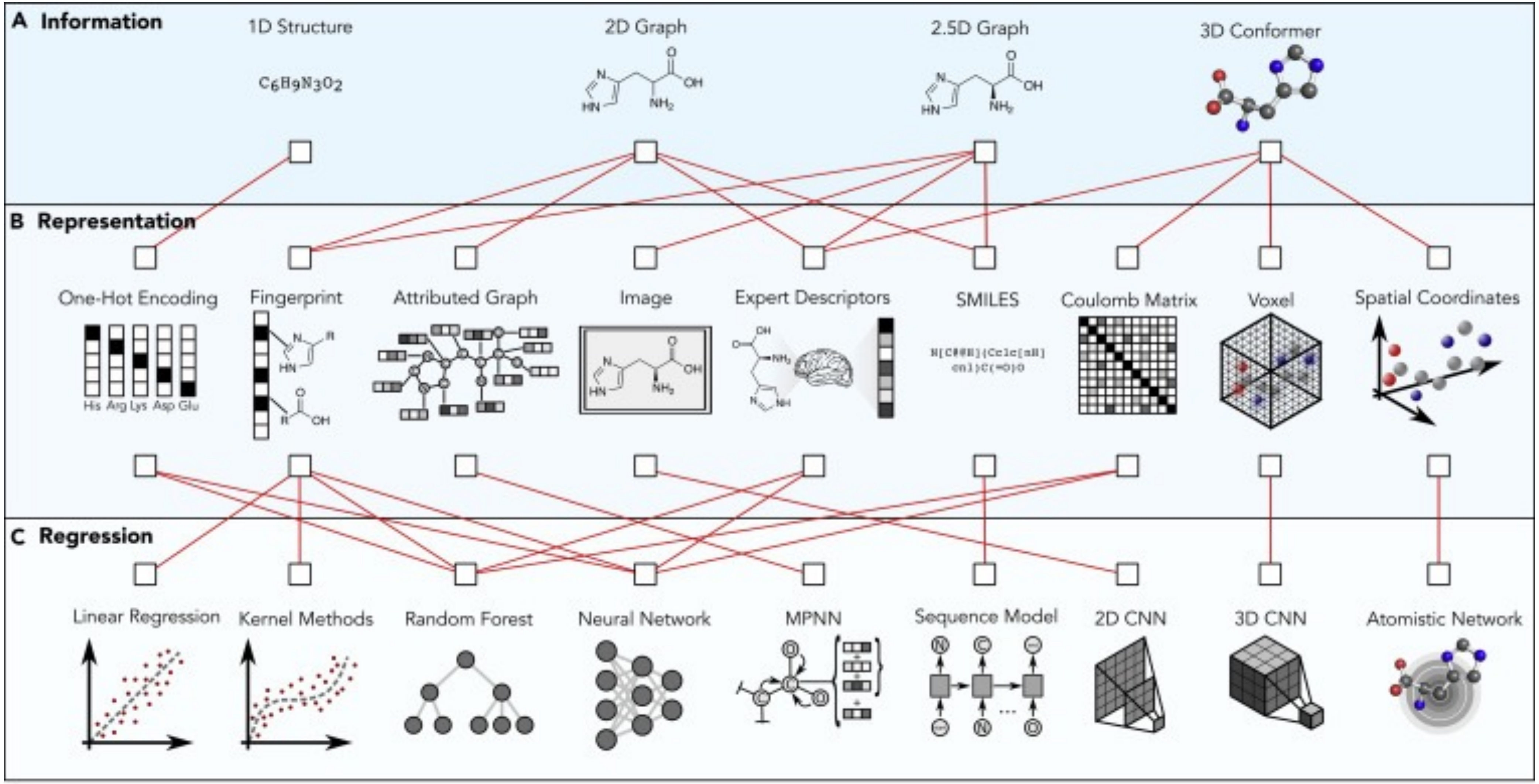
Bias into Engineered descriptors?

Example of solvent encoding



Ball, M.; Horvath, D.; Kogej, T.; Kabeshov, M.; Varnek, A. Predicting Reaction Conditions: A Data-Driven Perspective. *Chem. Sci.* **2025**, 16 (38), 17523–17541.

Molecular Representations: Outlook

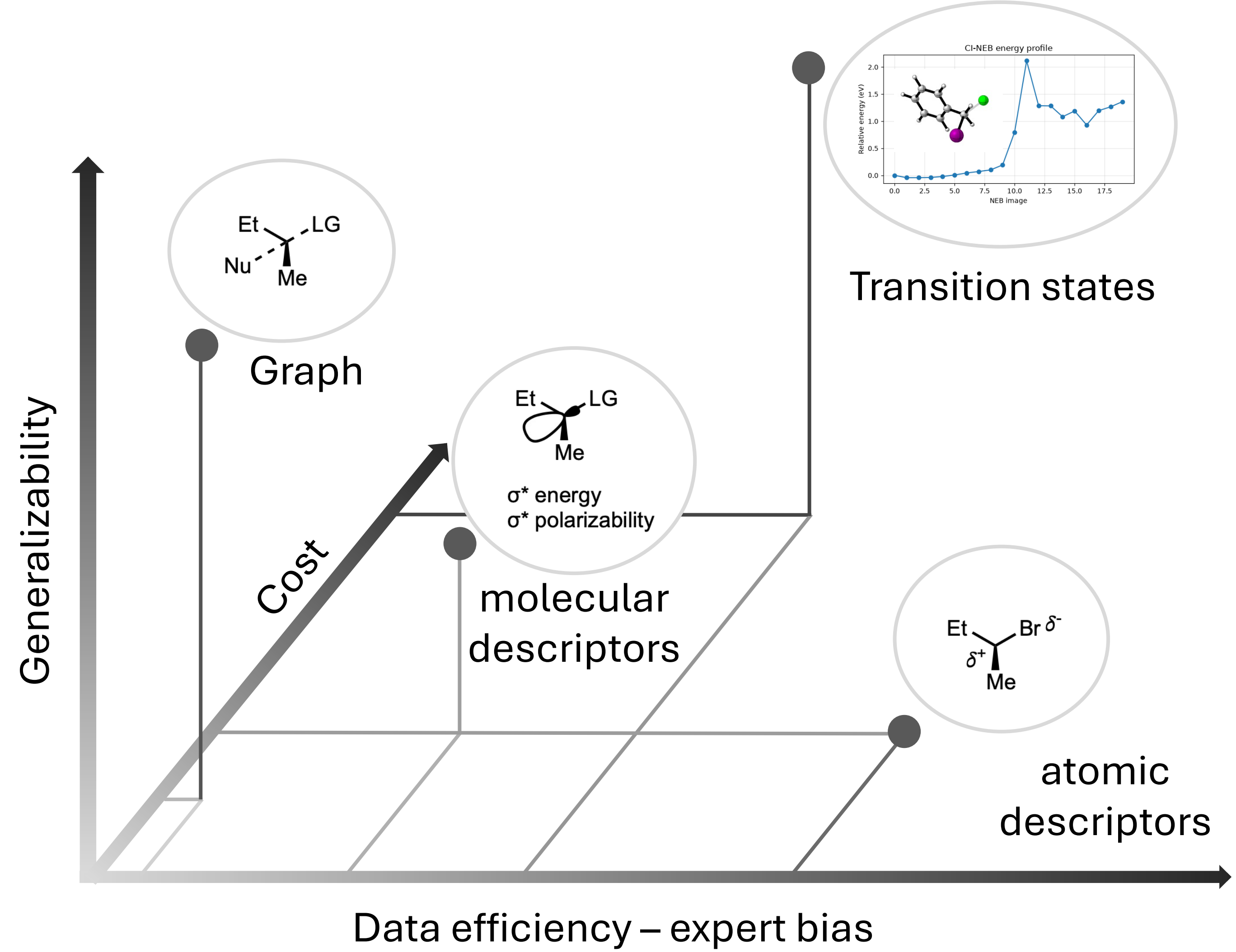


Pattanaik, L.; Coley, C. W. *Chem* **2020**, 6 (6), 1204–1207.

Plan

- Evolution of "data-driven" chemistry
- Overview of existing representations
- **Importance of the choice of the representation**
- Data preprocessing: practical considerations
- Data visualization

Expert-Crafter Descriptors: Precision/Generalizability Trade-off



Plan

- Evolution of “data-driven” chemistry
- Overview of existing representations
- Importance of the choice of the representation
- **Data preprocessing: practical considerations**
- Data visualization

Multiple feature preprocessing

- **Standardization**
- **Removing zero-variance features**
- **Removing correlated features**
- **Data visualization: understanding local and global data structure/potentially removing outliers**

Plan

- Evolution of “data-driven” chemistry
- Overview of existing representations
- Importance of the choice of the representation
- Data preprocessing: practical considerations
- **Data visualization**

Principal Component Analysis (PCA)

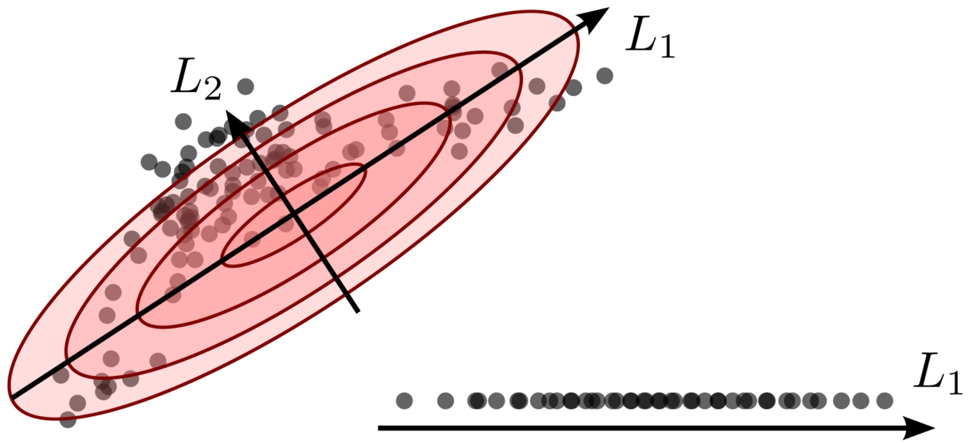
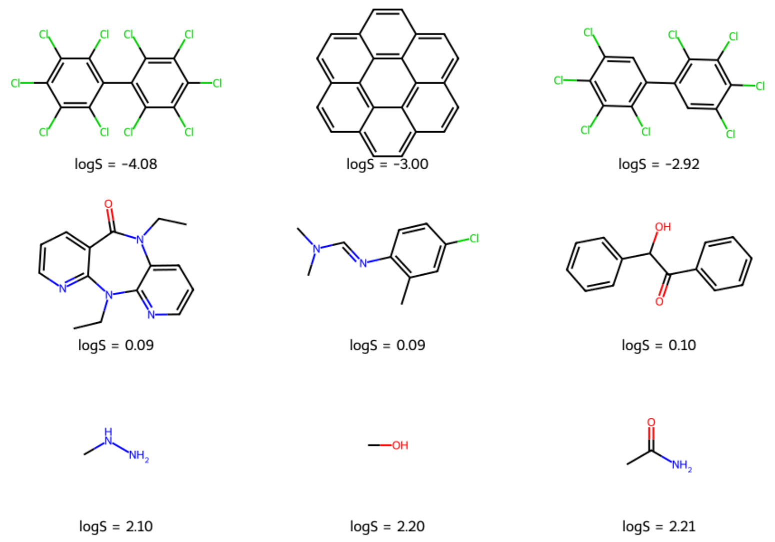


Figure from: https://github.com/BingqingCheng/ML-in-chemistry-101/blob/master/slides/l5.pdf

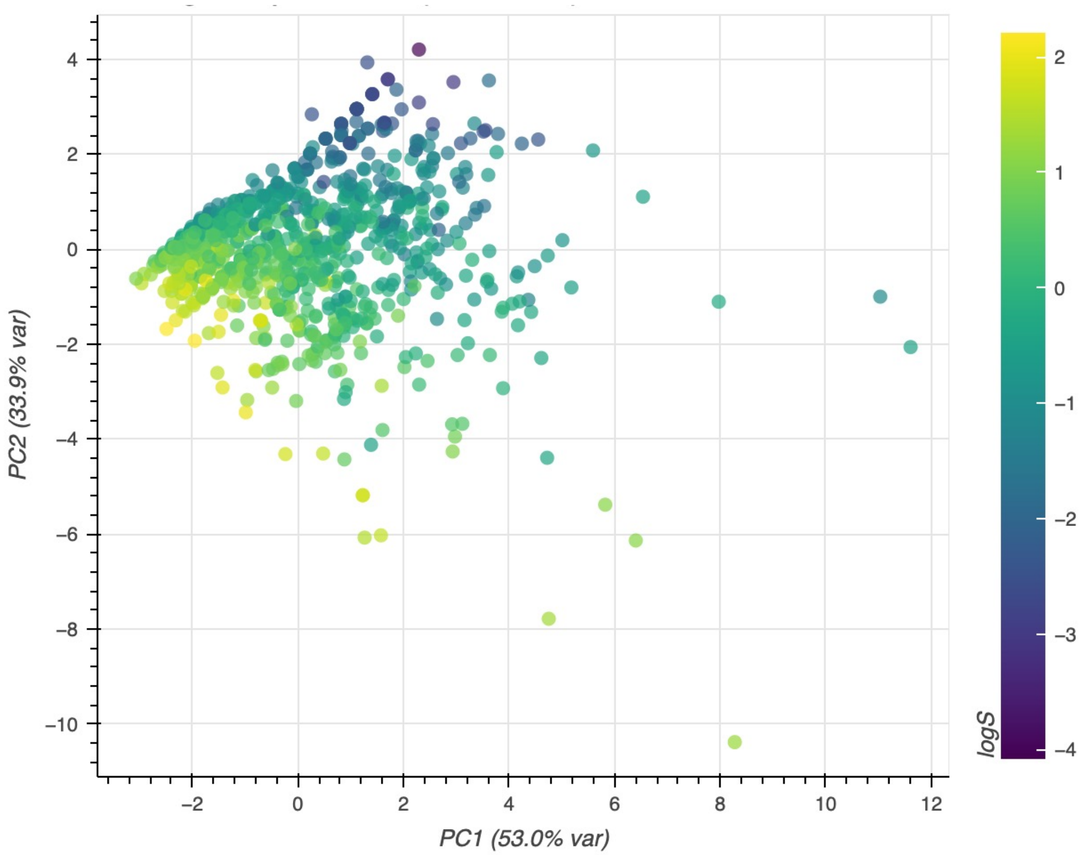
PCA: an example

Example molecules



Molecular descriptors used:
MolWt, TPSA, #HeavyAtoms, #Ring, MolLopP, #HDonors

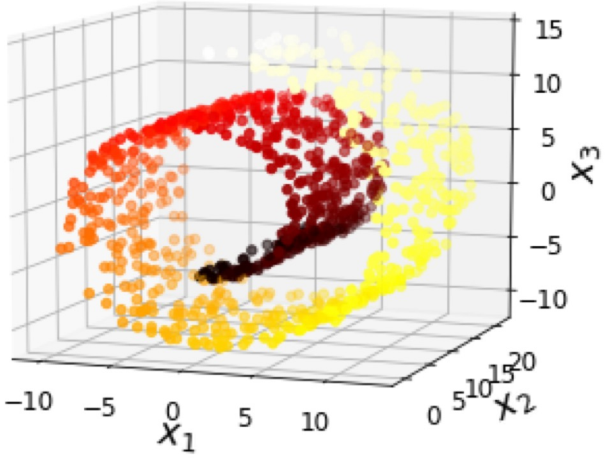
PCA projection (PC1 and PC2)

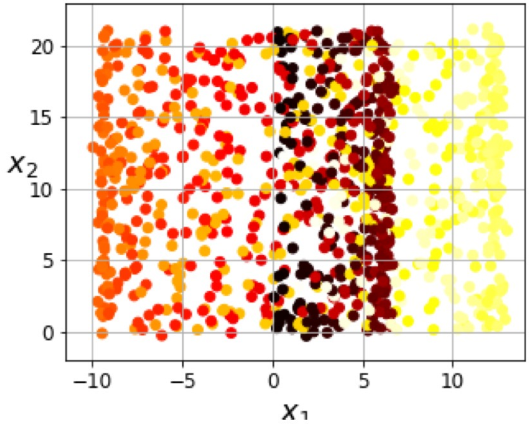


Total variance explained = 86.9%

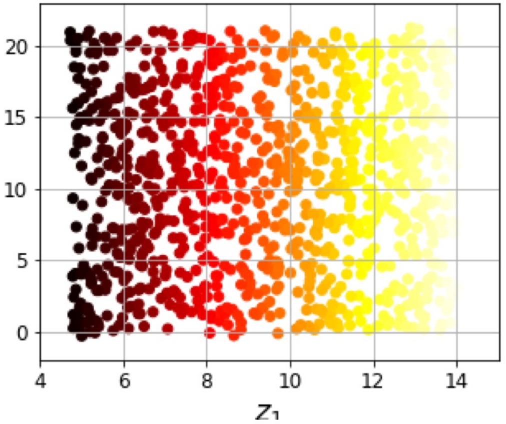
Delaney, J. S. *J. Chem. Inf. Comput. Sci.* **2004**, 44 (3), 1000–1005.

PCA limitations





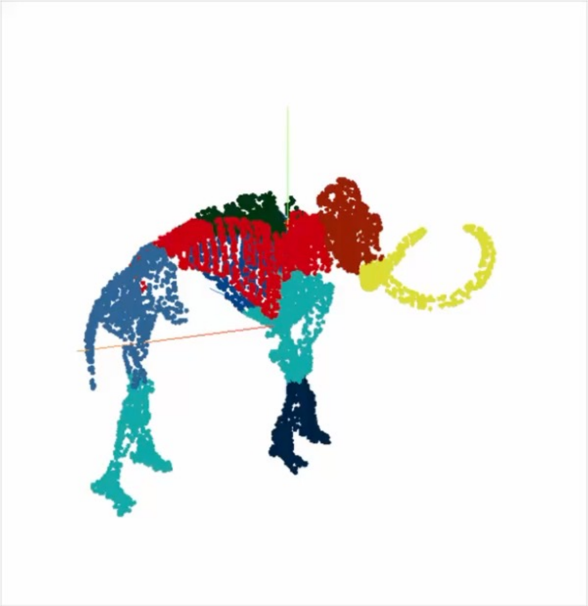
What you get with PCA



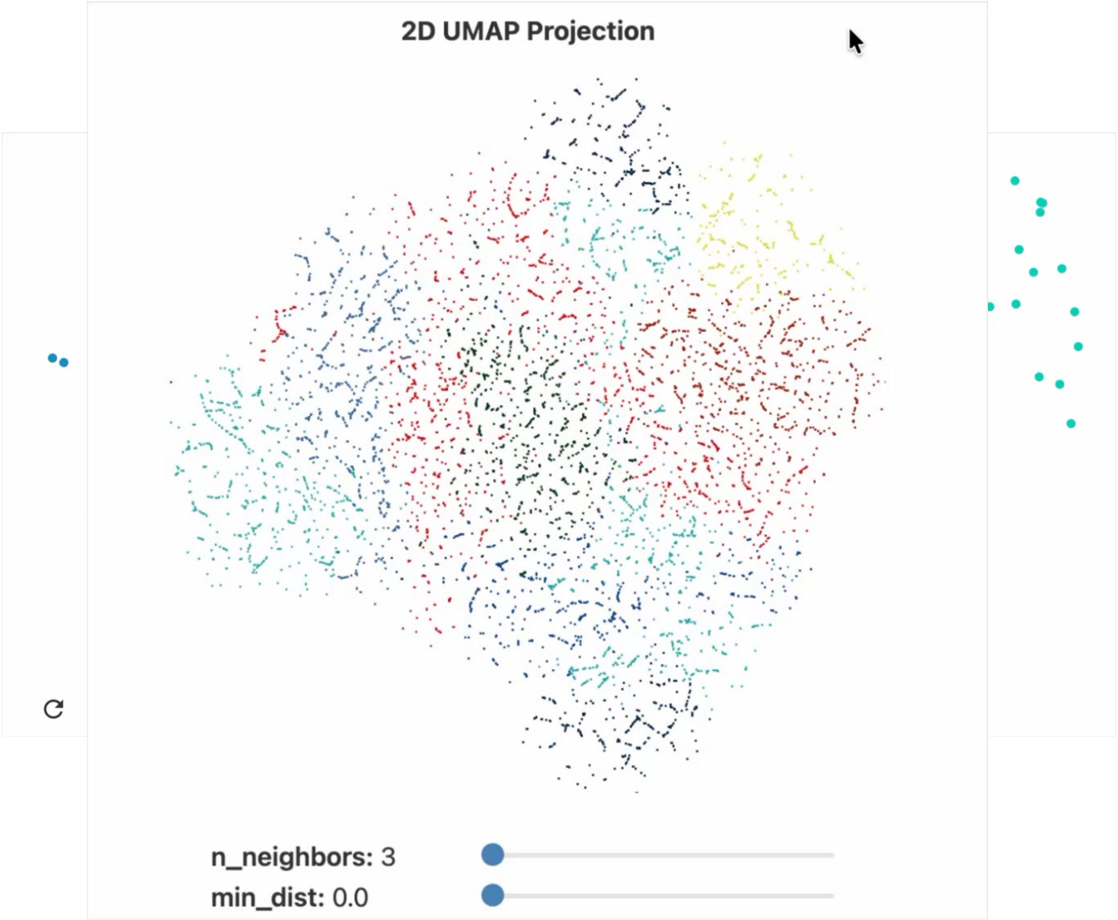
What you want

Figure from: https://github.com/BingqingCheng/ML-in-chemistry-101/blob/master/slides/l5.pdf

Beyond PCA for Dimensionality Reduction



How to preserve local and global structure while reducing dimension?

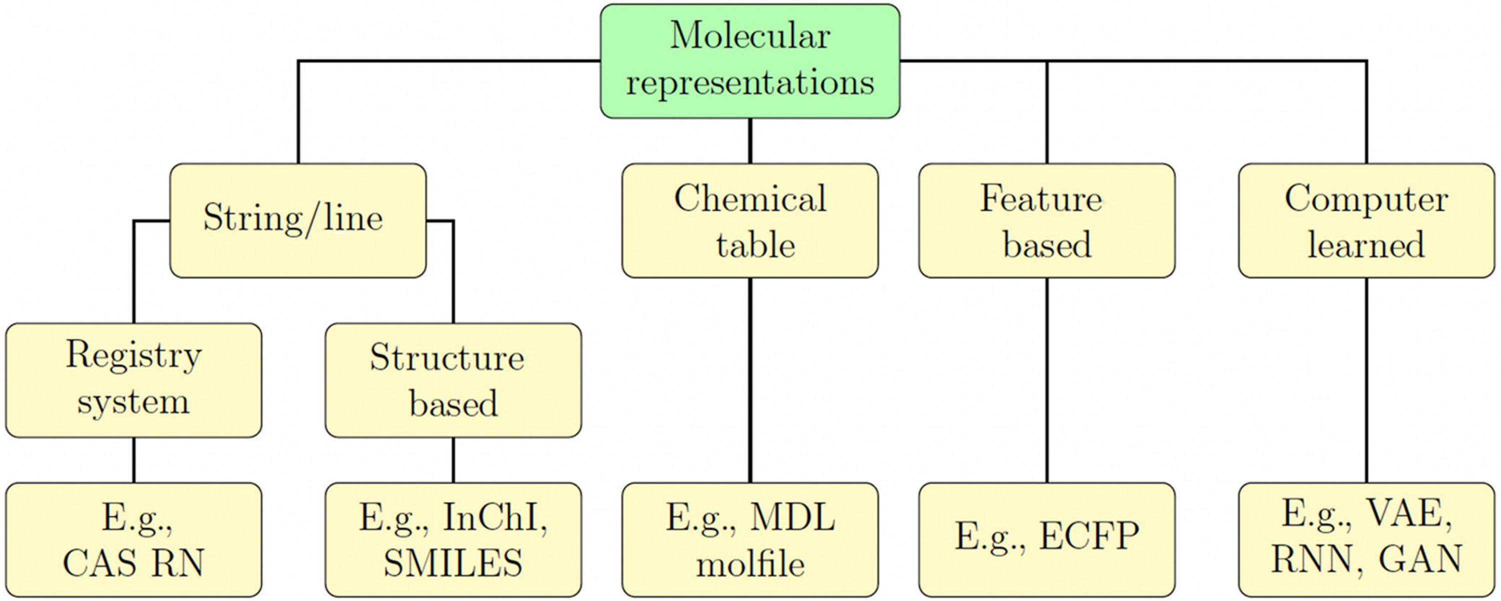


1. Hyperparameters really matter.
2. Cluster sizes in a UMAP plot mean nothing.
3. Distances between clusters might not mean anything.
4. Random noise doesn't always look random.
5. You may need more than one plot

Conclusion: Do not overinterpret!!!

https://pair-code.github.io/understanding-umap/

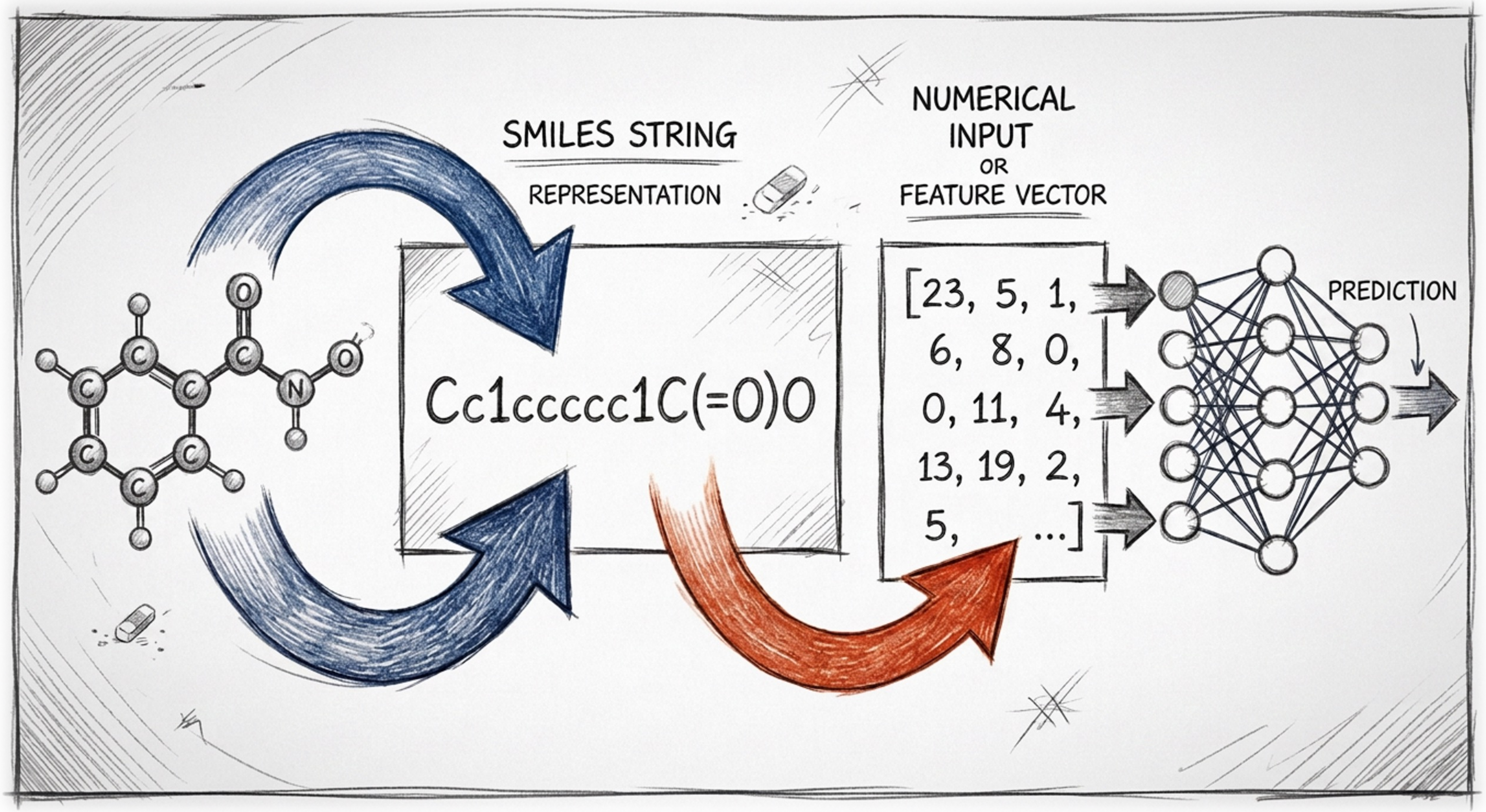
Future Lectures



- Graphs Neural Networks
- Variational Auto-Encoders
- Transformers

Wigh, D. S.; Goodman, J. M.; Lapkin, A. A. *WIREs Comput Mol Sci* **2022**, 12 (5).

Questions?



SMILES STRING
REPRESENTATION
NUMERICAL
INPUT
OR
FEATURE VECTOR
Cc1ccccc1C(=O)O
[23, 5, 1,
6, 8, 0,
0, 11, 4,
13, 19, 2,
5, ...]
PREDICTION